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MAKING SOME PROGRESS ON THE PROGRESSIVE ASPECT *ZÀI* IN MANDARIN CHINESE

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ABSTRACT

Taking into account data hitherto neglected, this article examines the progressive aspect *zài* 在 in Mandarin Chinese and related issues. We argue in favour of the constituency [_{AspP} *zài* 在 [_{V-zhe} 着]] and demonstrate important syntactic and semantic differences between the progressive aspect *zài* 在 and the durative aspect *-zhe* 着. Importantly, unlike ‘*zài* 在 V’, ‘V-*zhe* 着’ is only acceptable as the sole predicate in a root sentence when interpretable as a (derived) state. Otherwise, the presence of the progressive aspect *zài* 在 is required: ‘*zài* 在 V-*zhe* 着’. This crucial role of *zài* 在 is often obscured by its phonological fusion (*haplology*) with the preposition *zài* 在 ‘at’ heading an immediately following prepositional phrase: NP [_{AspP} *zài* 在 [_{VP} [_{PP} *zài* 在 NP] [_{V-zhe} 着 NP]]] => NP *ZÀI* 在 NP V-*zhe* 着 NP. The presence/absence of an underlying progressive aspect *zài* 在 (having fused with the preposition *zài* 在) is reflected in the choice of negation. Since the progressive aspect *zài* 在 must be negated by *méi* 没, not *bù* 不, it follows that no underlying aspectual *zài* 在 is present when it is *bù* 不 that negates a VP with a *zài* 在 PP adjunct:

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NP *bù* 不 [_{VP} [_{PP} *zài* 在 NP] VP]. By contrast, when there is an underlying aspectual *zài* 在, the negation *méi* 没 is required, resulting in the surface sequence ‘NP *méi* 没 *zài* 在 NP VP’. Finally, given that a subset of auxiliaries, some stative verbs as well as state-denoting ‘V-*zhe* 着’ are negated by *méi* 没, negation of ‘*zài* 在 VP’ by *méi* 没 is compatible with the status of ‘*zài* 在 VP’ as a state, hence able to be true at a moment.

KEYWORDS

Durative aspect -*zhe* 着 Haplology Negation *bù* 不 vs *méi* 没 Finiteness

1. INTRODUCTION

When wanting to examine the progressive aspect in Mandarin Chinese, one first needs to decide whether besides the clear case of the progressive aspectual auxiliary *zài* 在 in ‘*zài* 在 VP’, the sequence *zhèng* 正 (‘just’) *zài* 在 (‘PROGR’) is likewise to be included as another realization of the progressive aspect. While in most of the Chinese literature, this sequence is presented as an adverb, i.e., as one word, *zhèngzài* 正在 ‘just’, Sun Hongyuan (2014) and Martin et al. (2020) treat *zhèngzài* 正在 as a bisyllabic word variant of the progressive aspect *zài* 在. As we will see, none of the above scenarios is correct. Instead, the sequence ‘*zhèng* 正 *zài* 在 VP’ is to be analysed as the adverb *zhèng* 正 ‘just, right then’ preceding the progressive aspect head *zài* 在, which in turn selects the following verbal projection as its complement:

- (1) [_{AspP} [_{Adv} *zhèng* 正 ‘just’] [_{AspP} *zài* 在 [_{VP} VP]]]

This analysis not only allows us to correctly predict the cases where *zhèng* 正 ‘just, right then’ and progressive *zài* 在 can only occur on their own, but also to obtain the correct distribution and well-formedness conditions for the progressive aspect *zài* 在 itself.

The article is organized as follows. Section 2 provides evidence for our analysis in (1) where the adverb *zhèng* 正 ‘just, right then’ precedes and modifies the verbal projection headed by the progressive aspect *zài* 在. Section 3 turns to the structural analysis of the sequence ‘*zài* 在 V-*zhe* 着’ and the syntactic and semantic differences between the progressive aspect

zài 在 and the durative aspect *-zhe* 着. Section 4 addresses the phonological fusion of progressive *zài* 在 with the homophonous preposition *zài* 在 heading an immediately following PP, thus resulting in a single overt instance of *zài* 在: NP [*zài* 在 [_{PP} *zài* 在 NP] VP] => NP ZÀI 在 NP VP. This important phenomenon (*haplology*, cf. Chen Chung-yu 1978) is neglected in the majority of studies on the progressive *zài* 在 and leads to a distorted picture of its distribution. Section 5 tackles the difficult issue of negation. While there is a consensus that the projection headed by aspectual *zài* 在 must be negated by *méi* 没, not by *bù* 不, the implications of this requirement are interpreted differently in the literature and remain controversial. Importantly, the negation *méi* 没 will be shown to be compatible with (derived) states. Section 6 concludes the article.

The appendix (available at: <https://jcl.fah.um.edu.mo/wp-content/uploads/2023/11/appendix-formatted-26dec2022-updated.pdf>) discusses data from corpora and informant questionnaires indicating the existence of a small emerging group of speakers for whom *zhèngzài* 正在 is an adverb, i.e., a single word ‘just, right then’, as evidenced by the acceptability of the sequence ‘adverb *zhèngzài* 正在 + aspectual *zài* 在 + VP’. Importantly, these speakers (henceforth referred to as the “1-word *zhèngzài* 正在 speakers”) show the same properties for the aspectual *zài* 在, such as the haplology in the presence of an immediately adjacent preposition *zài* 在, leading to a single overt instance of *zài* 在 ‘NP *zhèngzài* 正在 [ZÀI 在 NP VP]’ and thereby nicely confirm this phenomenon.

2. IN FAVOUR OF ‘ADVERB *ZHÈNG* 正 ‘JUST’+ ASPECTUAL HEAD *ZÀI* 在’

If the sequence ‘*zhèng* 正 *zài* 在’ were indeed an adverb, i.e., a single word, hence to be noted as *zhèngzài* 正在 in the *Pinyin* transliteration, on a par with e.g., the adjunct NP *xiànzài* 现在 ‘now’, we would predict the well-formedness of the sequence ‘*zhèngzài* 正在 + aspectual *zài* 在’, in parallel to the sequence ‘*xiànzài* 现在 (now) + aspectual *zài* 在’ (cf. [2]). This is, however, not the case, ‘*zhèngzài* 正在 *zài* 在 VP’ being rejected (cf. [3]) (except for the emerging 1-word *zhèngzài* 正在 group presented in the appendix).

- (2) 他 现在 在 做 作业。
 Tā xiànzài zài zuò zuòyè.¹
 3SG now PROGR make homework
 ‘He’s doing his homework now.’
- (3) *他 正在 在 做 作业。
 *Tā zhèngzài zài zuò zuòyè.
 3SG zhengzai PROGR make homework

The majority of the data in favour of [_{Adv} zhèng 正] [_{AspP} zài 在 VP] to be provided below can in fact be found in the rich Chinese literature on these items, even though zhèng 正 ‘just, right then’, zài 在 ‘progressive aspect’ and zhèngzài 正在 are all assigned adverbial status here. (But cf. Zhang Huinü [2002] presented at the end of the next section for the analysis of zhèng 正 zài 在 as two words, not one).

2.1 Cases Where Either Zhèng 正 ‘Just’ or Aspectual Zài 在 Are Acceptable, But Not Their Combination *Zhèng 正 zài 在

One piece of evidence against the word status of zhèngzài 正在 and in favour of our analysis ‘adverb zhèng 正 + aspectual head zài 在 + VP’ are cases where either zhèng 正 ‘just, right then’ on its own (cf. [4] – [6]) or the aspect zài 在 on its own (cf. [7] – [9]) are acceptable, but where their combination in the sequence zhèng 正 zài 在 is excluded.

Let us start with illustrating the first case where only zhèng 正 ‘just, right then’ can precede the VP, to the exclusion of the aspect zài 在:

- (4) a. 你 来 得 真 巧, 我 正 (*在) 要 找 你。
 Nǐ lái de zhēn qiǎo, wǒ zhèng (*zài) yào zhǎo nǐ.²
 2SG come DE really opportune 1SG just PROGR want search 2SG
 ‘You’ve come at the right time, just as I wanted to go see you.’
- b. *你 来 得 真 巧, 我 在 要 找 你。
 *Nǐ lái de zhēnqiǎo, wǒ zài yào zhǎo nǐ.
 2SG come DE really opportune 1SG PROGR want search 2SG
- (5) a. 正 (*在) 出 门, 车 就 来 了。
 Zhèng (*zài) chū mén, chē jiù lái-le.
 just PROGR exit door bus then come-PERF
 ‘Just when he was coming out of the house, the bus arrived.’
 (based on Lü Shuxiang 2000: 670)

- b. *在 出 门, 车 就 来 了。
 *Zài chū mén, chē jiù lái -le.
 PROGR exit door bus then come-PERF
- (6) a. 我 中 午 回 到 家 正 (*在) 碰 上 老 李。
 Wǒ zhōngwǔ huí -dào jiā zhèng (*zài) pèngshàng Lǎo Lǐ.
 1SG noon return-arrive home just PROGR run.into Lao Li
 'Just when I came home at noon, I ran into Lao Li.'
- b. 他 正 (*在) 坐 在 沙 发 上 看 书。
 Tā zhèng (*zài) zuò-zài shāfā shàng kànshū.
 3SG just PROGR sit-be.at sofa on read book
 'He was just sitting on the sofa and reading.'
 (Zhang Huinü 2002: 63, [14]–[15], slightly modified)

The adverb *zhèng* 正 'just, right then' can coincide with the speech time as in (4a), refer to an implicit point in time provided by the context (cf. [6b]) or to a point in time explicitly mentioned as in (5a) ('just as he was coming out of the the house') and (6a) ('the moment when I came home at noon').

The progressive aspect *zài* 在 is excluded in the examples above because it is incompatible with auxiliaries such as *yào* 要 'want' (cf. [4]), with telic predicates such as *chū mén* 出门 'exit door' = 'leave' (cf. [5]) and stative predicates such as *zuò-zài* 坐在 'sit-be.at', respectively (cf. [6b]).

We now turn to the cases where only the aspectual *zài* 在 without the adverb *zhèng* 正 is allowed: '(**zhèng* 正) *zài* 在 VP':

- (7) a. 太 阳 每 天 都 (*正) 在 升 起 落 下。
 Tàiyáng měi tiān dōu (*zhèng) zài shēngqǐ luòxià.
 sun every day all just PROGR rise set
 'The sun is rising and setting every day.'
- b. 她 每 天 都 (*正) 在 学 汉 语。
 Tā měi tiān dōu (*zhèng) zài xué hànyǔ.
 3SG every day all just PROGR learn Chinese
 'She is studying Chinese every day.'
- (8) 大 姐 什 么 时 候 都 (*正) 在 认 真 工 作。
 Dàjiě shénme shíhou dōu (*zhèng) zài rènzhēn gōngzuò.
 e.sister what time all just PROGR conscientious work
 'Elder sister is conscientiously working all the time.'
- (9) a. 从 上 午 九 点 直 到 晚 上 九 点,
 Cóng shàngwǔ jiǔ diǎn zhídào wǎnshàng jiǔ diǎn,
 from morning 9 o'clock until evening 9 o'clock
- (Zhang Huinü 2002: 63, [16])

除了 匆匆 吃 饭, 她 始终 (*正) 在 写。
 chúle cōngcōng chī fàn, tā shǐzhōng (*zhèng) zài xiě.
 except hurriedly eat food 3SG throughout just PROGR write
 ‘From 9 am to 9 pm, except for hurriedly eating something,
 she has been writing all the time.’

(Guo Fenglan 1998: 37, [5], slightly modified)

- b. 她 [PP 从 一点 起] 一直 都 (*正) 在 打 字。
 Tā [PP cóng yīdiǎn qǐ] yīzhí dōu (*zhèng) zài dǎ zì.
 3SG from 1 o'clock on non-stop all just PROGR strike character
 ‘She has been continuously typing from 1 o'clock on.’

The adverb *zhèng* 正 ‘just, right then’ is excluded in (7)–(9), because its semantics is incompatible with the universal quantifier adverb *dōu* 都 ‘all’. While in (7) and (8), *dōu* 都 ‘all’ quantifies over the temporal expressions *měi tiān* 每天 ‘every day’ and *shénme shíhòu* 什么时候 ‘any time’ and the associated habitual eventualities, in (9a) and (9b), *shǐzhōng* 始终 ‘throughout’ and *dōu* 都 ‘all’ refer to the time span (‘from 9 am to 9 pm’ and ‘from 1 o'clock on’, respectively) with the Chinese equivalent of a “universal perfect” reading.

To sum up, an analysis treating *zhèngzài* 正在 as one word, more precisely as an adverb, would be unable to predict the data above, where it is totally excluded to replace the (genuine) adverb *zhèng* 正 ‘just, right then’ by the alleged adverbs *zhèngzài* 正在 or *zài* 在. Furthermore, the dominant adverbial analysis of *zài* 在 in most of the Chinese literature (cf. a.o. Lü Shuxiang 2000, Fan Jiyan 1982, Guo Fenglan 1998) completely glosses over the fact that unlike the adverb *zhèng* 正 ‘just, right then’, *zài* 在 as an aspectual head selects its complement and therefore imposes restrictions on the aktionsart of the following verbal projection, as evidenced by examples (4)–(6) above.³ Sun Hongyuan’s (2014) and Martin et al.’s (2020) analysis of *zhèngzài* 正在 as an aspectual head, hence on a par with the progressive *zài* 在, likewise encounters problems and incorrectly predicts the acceptability of *zhèngzài* 正在 in (7)–(9) above. It thereby blurs the role of *zài* 在 in habitual eventualities (cf. [7–8]) and in the Chinese equivalents of the universal perfect (cf. [9a–b]).⁴

Before going on to the next section, we would like to mention Zhang Huinü (2002), who to our knowledge is the only one to argue against *zhèngzài* 正在 as one word. Importantly, she explicitly states that *zhèng* 正

‘just, right then’ and *zài* 在 do not even form a constituent, as indicated in the bracketing below:

- (10) a. 老栓 正 [在 [专心 走路]],
 Lǎo Shuān zhèng [zài [zhuānxīn zǒu lù]],
 Lao Shuan just PROGR concentrate walk road
 忽然 吃了一惊。
 hūrán chī-le yī jīng.
 suddenly eat-PERF 1 shock
 ‘Lao Shuan was just concentrating on walking (when) he suddenly startled.’
 (Zhang Huinü 2002: 64, [42]–[44], slightly modified; our glosses and translation)
- b. 我们 正 [在 [做 我们 的前人
 Wǒmen zhèng [zài [zuò [NP women de qiánrén
 1PL just PROGR do 1PL SUB predecessor
 从来 没 做过 的 [...] 事业。
 cónglái méi zuò-guo de [...] shìyè]]].
 ever NEG do -EXP SUB work
 ‘We are just doing the things our predecessors have never done.’
- c. 雅座间 已经 坐了 一些人,
 Yǎzuòjiān yǐjīng zuò-le yīxiē rén,
 lounge already sit -PERF several people
 正 [在 [和 侃姐 起劲地 谈论
 zhèng [zài [hé Kǎn Jiě qǐjīnde tánlùn
 just PROGR with Kan Jie enthusiastically discuss
 法国 的 奶酪。
 fǎguó de nǎilào]]
 France SUB cheese
 ‘In the lounge had already sat down a number of people, and they were just enthusiastically discussing French cheese with Kǎn Jiě.’

However, elsewhere she adopts the current view of Chinese scholars and incorrectly considers both *zhèng* 正 ‘just, right then’ and the aspectual *zài* 在 as adverbs.

2.2 *Zhèng* 正 ‘Just, Right then’ as a VP-level Adverb

While there is no consensus on the categorial status of *zhèngzài* 正在, adverb or aspectual head (for those who treat it as one word), there is a general agreement that *zhèng* 正 ‘just, right then’ is an adverb. More

precisely, the data in the literature show it to be a low, i.e., VP-level adverb that must adjoin to, i.e., precede the verbal projection it modifies, i.e., in the case at hand the verbal projection headed by the aspectual *zài* 在.

Being a VP-level adverb itself, *zhèng* 正 ‘just, right then’ is mutually exclusive with other VP-level temporal adverbs such as *hái* 还 ‘still’, *réng* 仍 ‘still’, *yǐjīng* 已经 ‘already’, *yīzhí* 一直 ‘continuously’, *yòu* 又 ‘again’, whereas it can co-occur with, i.e., follow higher adverbs such as *dàgài* 大概 ‘approximately, probably’, *yě* 也 ‘also’, *dōu* 都 ‘all’ etc. (cf. [16], [17] below). *Zhèng* 正 is likewise compatible with manner adverbs, which as typical “ad-verbs” are closer to the VP than other co-occurring VP-level adverbs and hence appear below, i.e., to the right of *zhèng* 正 ‘just’:⁵

- (11) a. 他 正 在 考虑 问题。
Tā zhèng zài kǎolǜ wèntí.
3SG just PROGR ponder question
‘He’s just pondering the question.’
b. 他 一直 在 考虑 问题。
Tā yīzhí zài kǎolǜ wèntí.
3SG continuously PROGR ponder question
‘He has been continuously pondering the question.’
c. *他 {一直 /正}/ {正/ 一直} 在 考虑 问题。
*Tā {yīzhí /zhèng}/{zhèng/yīzhí} zài kǎolǜ wèntí.
3SG continuously /just just/ cont. PROGR ponder question
(based on Lü Shuxiang 2000: 672, point 4)

(11a–b) illustrates that both *zhèng* 正 ‘just, right then’ and *yīzhí* 一直 ‘continuously’ immediately precede ‘*zài* 在 VP’, but cannot co-occur, irrespective of their relative order (cf. [11c]). The same holds for the other VP-level adverbs in the examples below, where for simplicity sake only the unacceptable order with a following *zhèng* 正 ‘just, right then’ is indicated:

- (12) a. 司机 还 (*正) 在 唠叨。
Sījī hái (*zhèng) zài lāodao.⁶
driver still just PROGR chatter
‘The driver is/was still chattering.’
b. 我 发现 他 刚才 注视 我的 那 一眼
Wǒ fāxiàn tā gāngcái zhùshì wǒ de nà yī yǎn
1SG discover 3SG a.moment.ago gaze 1SG SUB that 1 eye

- 仍 (*正) 在 持续。
 réng (*zhèng) zài chíxù.
 still just PROGR continue
 ‘I discovered that that eye of hers that had gazed at me a moment ago still continued [to gaze].’
- c. 当时 我 已经 (*正) 在 胡思乱想。
 Dāngshí wǒ yǐjīng (*zhèng) zài húsi luànxiǎng.
 that.time 1SG already just PROGR imagine.things
 ‘I was already imagining things at that time.’
- d. 他们 又 (*正) 在 亲热地 交谈。
 Tāmen yòu (*zhèng) zài qīnrède jiāotán.
 3PL again just PROGR affectionately chat
 ‘They are/were again chatting cordially.’
 Zhang Huinü (2002: 63; [18], [22], [23], [39])

Note that in (12d) the verbal projection selected by the aspectual head *zài* 在 features a manner adverb (*qīnrède* 亲热地 ‘affectionately’), a data point that plays an important role for the correct analysis of the sequence ‘*zài* 在 V-*zhe* 着 (NP)’ discussed in section 3 below.

Furthermore, reviewer 2 provides example (13a) with the manner adverb between *zhèng* 正 ‘just, right then’ and the progressive aspect *zài* 在, thus providing further evidence against treating the sequence *zhèng* 正 *zài* 在 as one word:⁷

- (13) a. 他 正 目不转睛地 在 观察
 Tā zhèng mùbùzhuǎnjīngde zài guānchá
 3SG just intently PROGR watch
 仪表 读数 的 变化。
 yíbiǎo dúshǔ de biànhuà.
 meter reading SUB change
 ‘He is watching the changes in the meter reading intently.’
- b. 他 正在 目不转睛地 观察
 Tā zhèng zài mùbùzhuǎnjīngde guānchá
 3SG just PROGR intently watch
 仪表 读数 的 变化。
 yíbiǎo dúshǔ de biànhuà.
 meter reading SUB change
 ‘He is watching the changes in the meter reading intently.’

A quick informal survey with five native speakers from the North and two from the South showed their 100% acceptance for (13b) where the manner adverb directly adjoins to the VP. By contrast, for (13a), there were three “accept”, one clear “reject” and one “very marginal” for the Northern speakers, and two “accept” for the Southern speakers.

We also searched the BLU-BCC corpus (<http://bcc.blcu.edu.cn>) for manner adverbs ending in *-de* 地 between the adverb *zhèng* 正 ‘just, right then’ and the progressive aspect *zài* 在 and found 180 results. While in the majority of cases *zài* 在 was a (surface) preposition: ‘*zhèng* 正 manner-adv. [*zài* 在 NP] VP’ (cf. [14]), 43 results involved the progressive aspect *zài* 在 as in ‘*zhèng* 正 manner-adv. *zài* 在 VP’ (cf. [15]):

- (14) 成瑶 正 焦急地 [PP 在 宿舍 里] 等 他。
 Chéngyáo zhèng jiāojíde [PP zài sùshè lǐ] děng tā.
 Chengyao just anxiously at dormitory in wait 3SG
 ‘Cheng Yao is anxiously waiting for him in the dormitory.’
- (15) 网络 病毒 正 不断地 在 增加 和 改变。
 Wǎngluò bìngdú zhèng bùduànde zài zēngjiā hé gǎibiàn.
 internet virus just constantly PROGR increase and change
 ‘Internet viruses are constantly increasing and changing right now.’

Finally, the VP-level adverb *zhèng* 正 ‘just, right then’ can be preceded by sentence-level adverbs, e.g., *dàgài* 大概 ‘probably’, *yě* 也 ‘also’:

- (16) 这 时候 他 大概 正 在 准备 过节 呢。
 Zhè shíhòu tā dàgài zhèng zài zhǔnbèi guòjié ne.
 this moment 3SG probably just PROGR prepare holiday SFP
 ‘At this moment, he is probably just preparing for the celebration of the festival.’ (Guo Fenglan 1998: 36, [4], slightly changed)
- (17) 摇晃 着 身体 弹 着 吉他,
 Yáohuàng-zhe shēntǐ tán -zhe jítā,
 sway -DUR body pluck-DUR guitar
 他 也 正在 愉快地 唱。
 tā yě zhèng zài yúkuàide chàng.
 3SG also just PROGR happily sing
 ‘While she was swaying her body and playing the guitar, she was also singing happily.’
 (Zhang Huinü 2002: 64, [33], slightly changed)

In (17) *zhèng* 正 ‘just’ stresses the simultaneity of two eventualities. Importantly, the examples above with adverbs between the subject and the aspectual *zài* 在 invalidate Tsai Wei-Tien’s (2008: 678, 684) stipulation that *zài* 在 raises to T°.⁸ (T° in Chinese is always empty, cf. Ernst 1994).

3. CONSTITUENCY [*ZÀI* 在 [V-ZHE 着]], *[[*ZÀI* 在 V] ZHE]

Having established that *zhèng* 正 *zài* 在 in ‘*zhèng* 正 *zài* 在 VP’ is not one word, but the adverb *zhèng* 正 ‘just, right then’ preceding and modifying the projection headed by the progressive aspect head *zài* 在, we can now turn to other syntactic structures featuring *zài* 在, i.e., ‘*zài* 在 V-*zhe* 着’, where the verb is suffixed by the durative aspect *-zhe* 着:

- (18) 他 一直 在 听着 收音机。
Tā yīzhí zài tīng-zhe shōuyīnjī.
3SG continuously PROGR listen-DUR radio
‘He’s continuously listening to the radio.’
- (19) 雨 仍然 在 下着, 风 仍然 在 吹着。
Yǔ réngrán zài xià-zhe, fēng réngrán zài chuī-zhe.
rain still PROGR fall-DUR wind still PROGR blow-DUR
‘The rain is still falling, the wind is still blowing.’
- (20) 此刻 坐 在 车 上, 大家 都 在 打着 电话。
Cǐkè zuò-zài chē shàng, dàjiā dōu zài dǎ-zhe diànhuà
now sit-be.at train on everyone all PROGR strike-DUR phone
‘Sitting in the train right now, everybody is talking on the phone.’

While the analysis ‘[_{AspP} [_{Asp°} *zài* 在] [V-*zhe* 着]]’ where the aspectual head *zài* 在 selects ‘V-*zhe* 着’ is the obvious one, given the well-established strict right-branching syntax in the Chinese extended verbal projection (up to TP) (cf. C.-T. James Huang 1982 and later work), the opposite constituency: [_{AspP} [_{VP} [_{Adv} *zài* 在] VP] -*zhe* 着] (with *zài* 在 incorrectly assigned adverbial status) has nevertheless been proposed by Li Tianshu (2015: 49, [85]).⁹ However, in addition to the strict head-initial nature of the verbal projection, there exist further arguments against *-zhe* 着 selecting the ‘VP-plus-adverb *zài* 在’ sequence as its complement.

The most trivial one arises when the VP is spelt out as ‘verb + object’ and results in the incorrect order * [_{AspP} [_{VP} [_{Adv} *zài* 在] [_{VP} V NP]] -*zhe* 着]. This contrasts with the correct order predicted by our analysis

‘[_{AspP} [_{Asp°} *zài* 在] [[V-*zhe* 着] NP]]’ where *-zhe* 着 as a verb suffix immediately attaches to the verb, as illustrated in (18)–(20) above.

Even if for the sake of the argument we gloss over this problem and in addition adopt our analysis where *zài* 在 is an aspectual head, not an adverb as in Li Tianshu (2015), the acceptability of manner adverbs and adjunct XPs modifying the VP clearly demonstrate the phrasal status of *zài* 在’s complement: [_{AspP} *zài* 在 [_{VP} [_{adv} XP] [[V-*zhe* 着] NP]]]; it is thus not a potential legitimate host for the suffix *-zhe* 着, which requires a head V°:

- (21) a. 他 还 [_{AspP} 在 [[抽抽搭搭地] 哭泣着]].
 Tā hái [_{AspP} *zài* [[*chōuchōudādāde*] *kūqì-zhe*]].
 3SG still PROG sobbingly weep-DUR
 ‘He’s still weeping and sobbing.’
 (Guo Fenglan 1998: 43, [37], slightly modified)
- b. 他 [_{Asp} 在 [[_{PP} 和 朋友] 交谈着]].
 Tā [_{Asp} *zài* [[_{PP} *hé péngyǒu*] *jiāotán-zhe*]].
 3SG PROG with friend chat -DUR
 ‘He is chatting with friends.’

In Li Tianshu’s (2015) scenario, (21a–b) cannot be derived, because the verb suffix *-zhe* 着 would have to illegitimately combine with a phrase.

It is probably the *prima facie* impression that the same verbs can occur in the three structures: ‘*zài* 在 V’, ‘V-*zhe* 着’ and ‘*zài* 在 V-*zhe* 着’ that underlies the indecision concerning the correct constituency for the sequence ‘*zài* 在 V-*zhe* 着’, i.e., [*zài* 在 [V-*zhe* 着]] or [[*zài* 在 V]-*zhe* 着]. However, Li Tianshu (2015: 3) herself provides the crucial data (without following up on them) in favour of the structure [*zài* 在 [V-*zhe* 着]]. Referring to Hara (2009), she states that positional verbs such as *zhàn* 站 ‘stand’ etc. cannot be selected as complement of the progressive aspect *zài* 在, while this is possible when these verbs are suffixed with *-zhe* 着:¹⁰

- (22) 她 在 {站 *(着) / 躺 *(着) / 坐 *(着)}。
 Tā *zài* {*zhàn* *(-*zhe*) / *tǎng* *(-*zhe*) / *zuò* *(-*zhe*)}.
 3SG PROG stand -DUR / lie -DUR / sit -DUR
 ‘She is standing/lying/sitting.’

This nicely confirms our analysis in terms of [*zài* 在 [[V-*zhe* 着] NP]].

Other considerations beyond head directionality and selectional restrictions also argue in favour of *zài* 在 as selecting the verbal projection to its right. It has since long been observed in the literature that unlike ‘*zài* 在 V’, ‘V-*zhe* 着’ is not acceptable on its own as the predicate in a root sentence (cf. Chen Gang 1980, Kong Lingda 1994: 440; Huang Nansong 1994: 446; also cf. Djamouri and Paul 2018 and references therein):¹¹

- (23) a. *大家 夸 着 你。
 *Dàjiā kuā -zhe nǐ.
 everybody praise-DUR 2SG
 b. *他 正 写 着 信。
 *Tā zhèng xiě -zhe xìn.
 3SG just write-DUR letter

(Kong Lingda 1994: 440)

When ‘V-*zhe* 着’ is selected as complement by *zài* 在, however, (23a–b) become acceptable:¹²

- (24) a. 大家 在 夸 着 你。
 Dàjiā zài kuā -zhe nǐ.
 everybody PROGR praise-DUR 2SG
 ‘Everybody is praising you.’
 b. 他 正 在 写 着 信。
 Tā zhèng zài xiě -zhe xìn.
 3SG just PROGR write-DUR letter
 ‘He is just writing a letter.’

By contrast, when ‘V-*zhe* 着’ can be interpreted as a (derived) state, a reading reinforced by the presence of (manner) adverbs in (25a–b), it is acceptable as the sole predicate in a root sentence, as is well-known:¹³

- (25) a. 大雪 像 撕碎了 的 棉絮 飘 着。
 Dàxuě xiàng sīsuì-le de miánxù piāo-zhe.
 heavy.snow like tear -PERF SUB cotton.wool float-DUR
 ‘The snow flutters like torn cotton wool.’

(Chen Gang 1980: 25, slightly modified)

- b. 小明 像 根木桩似的 站着。
 Xiǎomíng xiànggēnmùzhuāngshide zhàn-zhe.
 Xiaoming as.straight.as.a.stilt stand-DUR
 ‘Xiaoming is standing as straight as a stilt.’

- c. 学生们 目不转睛地 看着 黑板。
 Xuéshēngmen mùbùzhuǎnjīngde kàn -zhe hēibǎn.
 students with.fixed.eyes look-DUR blackboard
 ‘The students are looking at the blackboard with utmost
 concentration.’

(Huang Nansong 1994: 445, [54–55])

This has to be seen against the background that in Chinese, only stative predicates (stative verbs and adjectives) can function as full predicates without any aspect marker in a root sentence, as already observed by Kong Lingda (1994: 436, III) for psychological predicates (e.g., *xǐhuān* 喜欢 ‘like’) and for equative sentences with the copula *shì* 是 ‘be’:

- (26) a. 她 喜欢 游泳。
 Tā xǐhuān yóuyóǒng.
 3SG like swim
 ‘She likes swimming.’
 b. 我 是 老师。
 Wǒ shì lǎoshī.
 1SG be teacher
 ‘I’m a teacher.’

By contrast, bare non-stative predicates are only acceptable under a habitual interpretation (cf. [28a]) or simply unacceptable (cf. [27a]) when the habitual interpretation is semantically implausible. For an episodic interpretation, they require an aspect marker, an auxiliary or negation; a temporal adverb on its own is not sufficient (cf. Sun Hongyuan 2014: 50).

- (27) a. *王 经理 吃 烤鸭。
 *Wáng jīnglǐ chī kǎoyā.
 Wang manager eat roast.duck
 b. 王 经理 经常 吃 烤鸭。
 Wáng jīnglǐ jīngcháng chī kǎoyā.
 Wang manager often eat roast.duck
 ‘Manager Wang often eats roast duck.’

(Huang Nansong 1994: 445, §3.3)

- (28) a. 她 打网球/ 抽 烟。
 Tā dǎ wǎngqiú/ chōu yān.
 3SG hit tennis / inhale smoke
 ‘She plays tennis/smokes.’ (Not: ‘She is playing tennis/smoking.’)

- b. 此时此刻 她 *(在) 打网球/ 抽 烟。
 Cǐ-shí-cǐ-kè tā *(zài) dǎ wǎngqiú/ chōu yān.
 this-time-this moment 3SG PROGR hit tennis / inhale smoke
 ‘Right now, she is playing tennis/smoking.’
- (29) a. 上 个 月 他 看 *(了) 《三 国 演 义》。
 Shàng ge yuè tā kàn*(-le) “sān guó yǎnyì”.
 last CL month 3SG see -PERF 3 kingdom romance
 ‘Last month, he read the *Romance of the three kingdoms*.’
 (Sun Hongyuan 2014: 50, [16b] slightly modified)
- b. 我 见 到 力 川 的 时 候, 他 *(在) 画 一 幅 画。
 Wǒ jiàndào Lichuān de shíhou, tā *(zài) huà yī fú huà.
 1SG see Lichuan SUB time 3SG PROGR paint 1 CL picture
 ‘When I saw Lichuan, she was painting a picture.’
 (Sun Hongyuan 2014: 254, [7b] slightly modified)

The temporal interpretation of bare predicates is thus a function of the predicate’s semantic type with a clear dichotomy between states, on the one hand, and eventive predicates (activity, achievement, accomplishment), on the other (cf. Sun Hongyuan 2014: 54–55).

If we now look at this generalization in the light of “assertability in a root context” as the most language-general and theory-neutral definition of *finiteness*, the judgements observed in (23)–(29) above all fall in place. (For reasons of space, we somewhat simplify here; finiteness-related issues are likewise examined in section 4 below. Also cf. Paul and Ramchand [2023] for extensive discussion.)

(23a–b) are unacceptable, because the durative aspect *-zhe* 着 is special among the aspect markers in Chinese insofar as it produces an open-ended activity without telos (cf. a.o. Wu Jiun-Shiung 2007). Accordingly, it does not “save” the assertion, unlike the “genuine” aspect marker *zài* 在.¹⁴ By contrast, when *-zhe* 着 creates a derived homogeneous state as in (25a–c), ‘V-*zhe* 着’ is acceptable as root predicate, on a par with the “lexical” stative in (26a–b) (also cf. Djamouri and Paul 2018). This is the reason why (30a) below is only acceptable under the interpretation ‘He has shoes on’, not ‘He is putting on shoes’. For the latter, *zài* 在 must be used (cf. Teng Shou-hsin 1973: 21; 1975: 128).

- (30) a. 他 穿着 鞋子。
 Tā chuān-zhe xiézi.
 3SG put.on-DUR shoe
 ‘He has shoes on.’
 b. 他 在 穿 鞋子。
 Tā zài chuān xiézi.
 3SG PROGR put.on shoe
 ‘He is putting on shoes.’

(Teng Shou-hsin 1973: 21, footnote 6)

These facts can only be explained when *zài* 在 selects ‘V-*zhe* 着’, and not the other way round, once again confirming the status of *zài* 在 as an aspectual head, not as adverb. For temporal adverbs alone cannot induce an episodic reading for bare activity predicates (cf. [28b], [29a–b]) above.¹⁵

The fundamental difference between temporal adverbs and aspect markers illustrated above ties in with Sun Hongyuan’s (2014: 44–45) generalization that they do not contribute in the same way to the computation of tense in a Chinese sentence. Accordingly, it is important to distinguish between the adverb *zhèng* 正 ‘just, right then’, on the one hand, and the progressive aspect head *zài* 在, on the other, in contrast to the literature where both are either classified as adverbs or as aspect markers, a uniform analysis unable to correctly predict the data.

4. HAPLOLOGY

As demonstrated above, ‘V-*zhe* 着’ is only acceptable as the sole predicate in a root sentence when encoding a derived homogeneous state (in contrast to an open-ended activity); otherwise, the presence of e.g., the progressive aspect *zài* 在 is required. This fact is often obscured by the invisibility of the aspectual *zài* 在 due to its phonological fusion (*haplology*) with the preposition *zài* 在 ‘at’ heading an immediately following PP. This well-documented phenomenon (cf. a.o. Chen Chung-yu 1978, Fan Jiyan 1982) is neglected, hence not controlled for in most studies on *-zhe* 着, as in the very first example provided for *-zhe* 着 by Lin Jo-wang (2002: 258):

- (31) a. 汽车 在 公路 上 飞奔着。
 Qìchē zài gōnglù shàng fēibēn-zhe.
 car at street on gallop-DUR
 'Cars are galloping on the street.'
 (Lin 2002: 258, [1a]; our glosses, tones added)

Lin Jo-wang (2002) fails to realize and therefore leaves uncommented the obligatory implicit presence of the aspectual *zài* 在 in (31a). Evidence for the latter becomes visible when we compare (31a) with (31b) and (31c):¹⁶

- b. *汽车 飞奔 着。
 *Qìchē fēibēn-zhe.
 car gallop-DUR
 c. 汽车 在 飞奔 着。
 Qìchē zài fēibēn-zhe.
 car PROGR gallop-DUR
 'Cars are galloping.'

(31b) is unacceptable, because *fēibēn-zhe* 着 'gallop-DUR' as an open-ended activity cannot constitute the sole predicate in a root sentence. However, when selected as the complement of the aspectual *zài* 在, the resulting sentence is well-formed (cf. [31c]). It follows that (31a) must likewise contain the aspectual *zài* 在 in order to license 'V-*zhe* 着'. The same contrast obtains in (32)–(33), where ZÀI in small caps represents the haplogy of the aspectual *zài* 在 with the preposition *zài* 在 'at'.

- (32) a. 她 在 洗澡间 吹 着 口哨。
 Tā ZÀI xǐzǎojiān chuī -zhe kǒushào.
 3SG PROGR+at bathroom blow-DUR whistle
 'She was whistling in the bathroom.'
 b. *她 吹 着 口哨。
 *Tā chuī -zhe kǒushào.
 3SG blow-DUR whistle
 c. 她 在 吹 着 口哨。
 Tā zài chuī -zhe kǒushào.
 3SG PROGR blow-DUR whistle
 'She was whistling.'
- (33) a. 他 在 外面 打 着 网球。
 Tā ZÀI wàimiàn dǎ-zhe wǎngqiú.
 3SG PROGR+at outside hit-DUR tennis

Furthermore, her example (35) below where in the presence of a preverbal *zài* 在-PP, *-zhe* 着 is obligatory for positional verbs such as *zuò* 坐 ‘sit’, *tǎng* 躺 ‘lie’ etc. can only be accounted for when positing an implicit aspectual *zài* 在:²⁰

- (35) 他 在 树 下 坐 *(着) / 躺 *(着)。
 Tā ZÀI shù xià zuò*(-zhe)/ tǎng*(-zhe).
 3SG PROGR+at tree under sit -DUR/ lie -DUR
 ‘He’s sitting/lying under the tree.’
 (Chen Chung-yu 1978: 91, [26a–b], glosses added)

Because the same observation holds for aspectual *zài* 在 on its own (cf. section 3 above): Bare positional verbs cannot be selected as complements by *zài* 在, whereas this is possible when they are suffixed by *-zhe* 着.

- (36) 他 在 坐 *(着) / 躺 *(着)。 (= [22] above)
 Tā zài zuò*(-zhe)/ tǎng*(-zhe).
 3SG PROGR sit -DUR/ lie -DUR
 ‘He’s sitting/lying.’

Note, though, that Chen Chung-yu (1978) has to assume “distant” haplology, because in her underlying structure, the preposition *zài* 在 and the aspectual *zài* 在 are not adjacent, but separated by the location NP:

- (37) NP [_{VP} [_{PlaceP} [_{Prep} *zài* 在] location NP] [_{VP} [_{Asp} *zài* 在] VP]]
 (Chen Chung-yu 1978: 82, [13a])

The relative hierarchy between aspect and PP in (37) is, however, incorrect, as rightly pointed out by Fan Jiyan (1982: 75). For in general the aspectual *zài* 在 precedes PPs, as illustrated below:

- (38) a. 他 [_{Asp} 在 [_{VP} [_{PP} 向 我] [_{VP} 招 手]]]。
 Tā [_{AspP} zài [_{VP} [_{PP} xiàng wǒ] [_{VP} zhāo shǒu]]].
 3SG PROGR towards 1SG beckon hand
 ‘He is beckoning me over.’
 (Fan Jiyan 1982: 75, [17]; glosses, brackets and translations added)
 b. 我 [_{AspP} 在 [_{VP} [_{PP} 对 你] [_{VP} 讲 话]]]。
 Wǒ [_{AspP} zài [_{VP} [_{PP} duì nǐ] [_{VP} jiǎng huà]]].
 1SG PROGR facing 2SG speak word

‘I’m speaking to you.’

- c. 雁 群 [AspP 在 [vP [PP 朝 南] [vP 飞翔]]]。
 Yàn -qún [AspP zài [vP [PP cháo nán] [vP fēixiáng]]]。
 wild.goose-group PROGR towards south fly
 ‘A flock of wild geese is flying towards the south.’

It follows that Fan Jiyan’s (1982: 75–76) structure in (39a) below is the correct input structure for the haplology; the PP headed by *zài* 在 ‘at’ is contained within the verbal projection selected as complement by the aspectual *zài* 在. Crucially, haplology is “local”, not “distant”, requiring adjacency of the aspectual *zài* 在 and the preposition *zài* 在:²¹

- (39) a. NP [AspP zài [vP [PP zài NP] [vP V NP]]] => NP ZÀI NP V NP
 b. 他 [AspP 在 [vP [PP 在 里屋] [vP 看 书]]]
 Tā [AspP zài [vP [PP zài lǐwū] [vP kàn shū]]]
 3SG PROGR at room see book
 => 他 在 里屋 [vP 看 书]
 Tā ZÀI lǐwū [vP kàn shū]
 3SG PROGR+at room see book
 ‘He’s reading in the back room.’
 (Fan Jiyan 1982: 75–76, [19a–c]; bracketing, glosses and translation added)

Fan Jiyan (1982: 78) furthermore points out that Chen Chung-yu’s (1978) assumption of haplology being able to apply at a “distance” would incorrectly predict the occurrence of only one *zài* 在 for sentences where the *zài* 在-PP occurs in the topic position preceding the subject:

- (40) [TopP [PP 在 教室 里],
 [TopP [PP Zài jiàoshì lǐ],
 at class.room in
 [TP 张 老师 正 *(在) 讲 几何代数]].
 [TP Zhāng lǎoshī zhèng *(zài) jiǎng jǐhé dàishù].
 Zhang teacher just PROGR talk set.algebra
 ‘In the class room, teacher Zhang is just talking about set algebra.’
- (41) [TopP [PP 在 门口], [TP 小黄 *(在) 痴痴地 笑]].
 [TopP [PP Zài ménkǒu], [TP Xiǎohuáng *(zài) chīchīde xiào]].
 at entrance Xiaohuang PROGR sneeringly laugh
 ‘In the entrance, Xiaohuang is sneering.’

Fan Jiyan (1982: 75) explicitly states that (40)–(41) with a topic *zài* 在-PP behave like “simple” sentences, i.e., sentences without any other occurrence of *zài* 在, and that accordingly the aspectual *zài* 在 – an adverb for Fan Jiyan (1982) – is obligatory for the progressive reading.²²

This clearly contrasts with the corresponding sentences (42)–(43) below, where the surface *zài* 在 PP *follows* the subject and where the aspectual *zài* 在 “fuses” with the preposition and is therefore invisible:

- (42) [TP 张 老师 [正 [在 教室 里 讲 几何代数]]]。
 [TP Zhāng lǎoshī [zhèng [ZÀI jiàoshì lǐ jiǎng jǐhé dàishù]].
 Zhang teacher just PROGR+at class.room in talk set.algebra
 ‘Teacher Zhang is just talking about set algebra in the class room.’
- (43) [TP 小黄 [在 门口 痴痴地 笑]]。
 [TP Xiǎohuáng [ZÀI ménkǒu chīchīde xiào]].
 Xiaohuang PROGR+at entrance sneeringly laugh
 ‘Xiaohuang is sneering in the entrance.’

The position of the adverb *zhèng* 正 ‘just, right then’ in (42) likewise argues in favour of Fan’s structure in (39a), because *zhèng* 正 as a low adverb must immediately adjoin to the (extended) verbal projection, here AspP.

The contrast between (40)–(41), on the one hand, and (42)–(43), on the other, brings us back to the issue of finiteness, i.e., assertability in a root sentence, and the crucial role aspect such as the progressive *zài* 在 plays here. Seen in this light, Chen Chung-yu’s (1978: 84) view cannot be correct that *zài* 在 functions as a “pure” preposition as soon as the verb is suffixed by an aspect marker, including durative *-zhe* 着.

On the contrary, the cases with perfective *-le* 了 and experiential *-guo* 过 (the latter not discussed by Chen Chung-yu) need to be distinguished from those involving V-*zhe* 着.

- (44) a. 他 (*在) 唱 了 两 首 歌。
 Tā (*zài) chàng-le liǎng shǒu gē.
 3SG PROGR sing -PERF 2 CL song
- b. 他 在 电视 上 唱 了 两 首 歌。
 Tā zài diànshì shàng chàng-le liǎng shǒu gē.
 3SG at TV on sing -PERF 2 CL song
 ‘He sang two songs (on TV).’

(Chen Chung-yu 1978: 82)

- (45) a. 她 (*在) 当 过 班长。
 Tā (*zài) dāng -guo bānzhǎng.
 3SG PROGR serve.as-EXP monitor
 b. 她 在 我们 学校 当 过 班长。
 Tā zài wǒmen xuéxiào dāng -guo bānzhǎng.
 3SG at 1PL school serve.as-EXP monitor
 ‘She served as class monitor (in our school).’

Since both perfective *-le* 了 and experiential *-guo* 过 are incompatible with the progressive aspect *zài* 在 (cf. [44a], [45a]), it follows that *zài* 在 in (44b) and (45b) is indeed the preposition ‘at’, not an instance of haplogy.

The case of ‘V-*zhe* 着’ is different, as illustrated by the contrast between (46a) and (46b–c), reported by Chen Chung-yu (1978: 83) herself:

- (46) a. *他 哭 着。
 *Tā kū -zhe.
 3SG cry-DUR
 b. 他 在 哭 着。
 Tā zài kū -zhe.
 3SG PROGR cry-DUR
 ‘He is crying.’
 c. 他 在 客厅 里 哭 着。
 Tā zài kètīng lǐ kū -zhe.
 3SG PROGR+at living.room in cry-DUR
 ‘He is crying in the living room.’

Given the obligatory presence of the aspectual *zài* 在 in (46b), it must be underlyingly present in (46c); accordingly, *zài* 在 in (46c) is the haplogy of aspectual *zài* 在 and the preposition *zài* 在 ‘at’, not the preposition.²³

Also recall from the discussion of (35) and (36) above (repeated in [47a–b] below) that both in sentences with the aspectual *zài* 在 and with a sentence-internal (surface) *zài* 在 PP, position verbs are only acceptable when suffixed by *-zhe* 着 :

- (47) a. 他 在 {坐 *(着) / 躺 *(着)}。
 Tā zài {zuò *(-zhe) / tǎng *(-zhe)}.
 3SG PROGR sit -DUR/ lie -DUR
 ‘He’s sitting/lying.’

- b. 他 在 树 下 {坐 *(着)/ 躺 *(着)}。
 Tā ZAI shù xià {zuò *(-zhe)/ tǎng *(-zhe)}.
 3SG PROGR+at tree under sit -DUR/ lie -DUR
 ‘He’s sitting/lying under the tree.’

Again, this can only be accounted for when acknowledging the obligatory underlying presence of the aspectual *zài* 在 in (47b) as well, invisible due to its haplology with the preposition *zài* 在 ‘at’.

Finally, Chen Chung-yu (1978) does not control for the absence/presence of the SFP *ne* 呢, a low complementizer, and therefore does not see that in the absence of *zài* 在, it is the SFP *ne* 呢 that licenses ‘V-*zhe* 着’ as the only predicate in a root sentence, when the latter cannot be interpreted as a (derived) state:

- (48) 他 上 着 课 *(呢)。
 Tā shàng-zhe kè *(ne).
 3SG attend-DUR class SFP
 ‘He’s in class now.’

(Chen Chung-yu 1978: 84, [16d]; glosses and parentheses added)

While Chen Chung-yu (1978) marks *ne* 呢 as optional, it is in fact required for the well-formedness of (48) as a root sentence. In general, the literature on *-zhe* 着 neglects the crucial role of *ne* 呢 for temporal anchoring and finiteness, notwithstanding the early observations by Chen Gang (1980), Zhu Dexi (1982, ch. 16) and Kong Lingda (1994) (cf. footnote 13 above) on *ne* 呢 in root sentences with ‘V-*zhe* 着’ as the only predicate. (Also cf. Djamouri and Paul 2018, Paul 2018 and references therein.)

To wrap up this section, when analysing the surface sequence ‘NP *zài* 在 NP VP’, it is absolutely necessary to control for the presence/absence of haplology, i.e., the potential underlying presence of the progressive aspect *zài* 在, invisible due to its phonological fusion with the preposition *zài* 在. Otherwise, the conditions under which aspect markers are required to ensure episodic readings in Chinese become inconsistent with the overall picture. The potential presence of haplology is likewise extremely important in order to understand negation of the progressive aspect *zài* 在, to be examined in the next section.

5. NEGATION

There is a general consensus in the literature that the AspP headed by the progressive *zài* 在 must be negated by *méi* 没, not by *bù* 不.²⁴

- (49) a. 你们 为什么 现在 {没/ *不} 在 做 作业?
 Nǐmen wèishénme xiànzài {méi /*bù} zài zuò zuòyè?
 2PL why now NEG/ NEG PROGR do homework
 ‘Why aren’t you doing your homework now?’
 b. 他 也许 / 大概 {没 /*不} 在 上 课。
 Tā yěxǔ / dàgài {méi /*bù} zài shàng kè.²⁵
 3SG perhaps/probably NEG / NEG PROGR attend class
 ‘He is perhaps/probably not attending classes.’

It goes beyond the scope of this article to fully address the complex and still not settled issue of the differences between these two types of negation in Mandarin Chinese. We therefore only briefly present some relevant observations from Ernst (1995) and Lin Jo-wang (2003) and then critically examine them in the light of the overall distribution of *bù* 不 and *méi* 没, including data not taken into account by them.

5.1 Two Types of Negation in Mandarin: *Bù* 不 vs *Méi* 没

Both Ernst (1995) and Lin Jo-wang (2003) account for the contrast between *bù* 不 and *méi* 没 in terms of different aspectual selection. They agree that stative predicates, i.e., adjectives and stative verbs, are negated with *bù* 不 only. Beyond this consensus, however, their basic premises differ, each of which has its own shortcomings.

According to Ernst (1995: 696) *bù* 不 requires “unbounded aspectual situations”, i.e., states, but also activity verbs for him. The latter are claimed to be in principle unbounded as well, because their possible endpoint is arbitrary (cf. footnote 25 on p. 698). As already seen in the preceding sections and to be confirmed in the discussion below, presenting activity verbs as patterning with stative predicates with respect to negation does not lead to correct predictions.

According to Lin Jo-wang (2003: 428), *bù* 不 exclusively selects states, and *méi* 没 non-states. More precisely, “*bù* 不 aspectually selects as its complement a stative situation *that requires no input of energy in order to obtain that situation*” (our emphasis). Unlike *bù* 不, *méi* 没 “aspectually

selects an event as its complement”, i.e., a non-state, a dynamic situation involving change over time and requiring input of energy (cf. Lin Jo-wang 2003: 428, [6a–b]). The stipulation “requires (no) input of energy” added in the definition of stative situations and events is necessary for Lin Jo-wang (2003) to be able to correctly predict that ‘zài 在 VP’- an event for him - requires the negation *méi* 没 and excludes *bù* 不:

- (50) 他 {没 /*不} 在 洗 澡。
 Tā {méi/ *bù} zài xǐ zǎo.
 3SG NEG/ NEG PROGR wash bath
 ‘He’s not taking a bath.’

(Lin Jo-wang 2003: 430, [9a], [10])

Lin Jo-wang (2003: 430) thus adopts Nordlander’s view (1997) that the situation described by a progressive sentence is *not* a state, but a process constituting “a number of consecutive phases” with “an ongoing supply of energy”, again in order to explain why the progressive aspect *zài* 在 is negated by *méi* 没. However, while Nordlander’s (1997) description might hold for the affirmative version of (50), i.e., ‘He is taking a bath.’ it is not clear at all whether it likewise applies to sentences such as (51) below:

- (51) 她 (没) 在 休息/ 睡/ 等。
 Ta (méi) zài xiūxi/ shuì/ děng.
 3SG NEG PROGR rest / sleep/ wait
 ‘She is (not) resting/sleeping/waiting.’

Be that as it may, Lin Jo-wang’s (2003) narrow dovetailed definition of stative situations as excluding any energy input and the ensuing distribution ‘*bù* 不 (only for states) vs *méi* 没 (only for non-states)’ in any case does not provide us with the correct picture.

Because some auxiliary verbs (e.g., *néng*(gòu) 能(够) ‘can’, *yào* 要 ‘want, will’, *kěnn* 肯 ‘be willing’, *gǎn* 敢 ‘dare’) likewise allow *méi* 没, in addition to the canonical negation with *bù* 不 holding for all auxiliaries (cf. Lü Shuxiang (2000: 384); also cf. Wang Canlong 2011).²⁶ This can hardly be accounted for in terms of presence vs absence of “energy input”.

Furthermore, some stative verbs (e.g., *zài* 在 ‘be at’) and derived states ‘V-*zhe* 着’ have *méi* 没 as their default negation, although in some cases *bù* 不 is likewise acceptable (cf. Lü Shuxiang 1985: 245–246):

- (52) 老李 在 家 吗? 没 /不 在 家。
 Lǎolǐ zài jiā ma? Méi/ bù zài jiā.
 Laoli be.at home SFP NEG/ NEG be.at home
 ‘Is Laoli at home? No, he isn’t.’
- (53) 我 现在 也 没 /不 闲 着。
 Wǒ xiànzài yě méi/ bù xián-zhe.
 1SG now also NEG/NEG idle-DUR
 ‘I’m not idle now.’
- (54) 我 看见 他 没 /不 躺 着, 就 叫 他 躺 下。
 Wǒ kànjiàn tā méi/ bù tǎng-zhe, jiù jiào tā tǎng-xià.
 1SG see 3SG NEG/NEG lie -DUR then tell 3SG lie -descend
 ‘I saw that he was not lying, so I told him to lie down.’

(54) shows that in combination with ‘V-*zhe* 着’, both *méi* 没 and *bù* 不 are compatible with a past tense interpretation.

Finally, as observed by Lin Jo-wang (2003: 431) himself, in the locative construction with V-*zhe* 着, only *méi* 没 is acceptable, to the exclusion of *bù* 不.²⁷

- (55) 墙 上 {没 /*不} 挂 着 画,
 Qiáng shàng {méi/*bù} guà -zhe huà,
 wall on NEG/ NEG hang-DUR picture
 也 没 挂 着 通告,
 yě méi guà -zhe tōnggào,
 also NEG hang-DUR announcement
 只 挂 着 一幅 挂历。
 zhǐ guà -zhe yī fú guàlì.
 only hang-DUR 1 CL wall.calendar
 ‘There are no pictures hanging on the wall, there are also no announcements hanging, there is just a wall calendar hanging on the wall.’

Lin Jo-wang (2003: 432) again refers to the (extralinguistic) fact that positional states such as lying, sitting or hanging require prior energy input. However, besides familiar examples from the literature (cf. [56a–b] below) where such a prior energy input is simply implausible due to our world

knowledge, there are well-known *linguistic* constraints holding for the locative construction with *V-zhe* 着, which directly challenge Lin Jo-wang's (2003) account.

- (56) a. 天 上 挂 着 一 轮 明 月。
 Tiān shàng guà -zhe yī lún míng yuè.
 sky on hang-DUR 1 CL bright moon
 'In the sky hangs a bright moon.'
 (Hu Wenzhe 1995: 106, [34], slightly changed)
- b. 细 细 的 枝 条 上 挂 着 绿 色 的 柿 子。
 Xìxì de zhītiáo shàng guà -zhe lǜsè de shìzi.
 thin SUB branch on hang-DUR green SUB persimmon
 'Green persimmons hung from thin branches.'
 'On thin branches are hanging green persimmons.'
 (Jaxontov 1988: 132, [78], our translations)
- (57) a. *桌子 上 {故意 / 慢慢地} 放 着 一 本 书。
 *Zhuōzi shàng {gùyì / mànmande} fang -zhe yī běn shū.
 table top on.purpose/ slowly put -DUR one CL book
 'A book was put on the table on purpose/slowly.'
 (Pan Haihua 1996: 430, [60]; his glosses and translation)
- b. 屋 里 {并 排 / *紧 张 地} 站 着 十 个 人。
 Wūlǐ {bìngpái / *jǐnzhāngde} zhàn -zhe shí ge rén.
 room side.by.side/ nervously stand-DUR 10 CL person
 'In the room are standing side by side/nervously ten people.'
- c. 北 门 上 (*主 动) 守 着 一 个 连。
 Běimén shàng (*zhǔdòng) shǒu -zhe yī ge lián.
 Northern.Gate on own.accord guard-DUR 1 CL company
 'At the Northern Gate several companies are keeping guard
 (of their own accord).'
- ([57b–c] based on Li Linding 1986: 75)

Since agent-oriented and activity related adverbs are excluded from the locative construction with 'V-zhe 着' (cf. [57a–c]), Paul, Lu and Lee (2020: 263) conclude that it conveys a state, rather than an agent-controlled activity or process. This directly challenges Lin-Jo-wang's (2003) claim that *méi* 没 only negates events, i.e., non-states.

To wrap-up, aspectual *zài* 在 is negated by *méi* 没. Importantly, in addition, *méi* 没 is also used as negation for (derived) states, including *V-zhe* 着, certain stative verbs as well as a subset of auxiliaries.²⁸ While the exact nature of these states as well as their sometimes observed

compatibility with both *méi* 没 and *bù* 不 must be left for further research, it is evident that Lin Jo-wang's (2003) association of negation by *méi* 没 with "input of energy", hence excluding states, is not on the right track, because inconsistent with the overall distribution of *méi* 没.

5.2 ‘{*Méi* 没/**bù* 不} *zài* 在 NP V-*zhe* 着’ and Haplology

Negation of the progressive aspect *zài* 在 with *méi* 没 also provides confirming evidence for the phenomenon of haplology discussed above.

Context: A phones B on her cell phone, i.e., A doesn't know what B is doing.

(58) a. A: 你 在 家 休息 吗?

A: Nǐ ZAI jiā xiūxi ma?

2SG PROGR+AT home rest SFP

‘Are you having a rest at home?’

b. B: (不,) 我 {没/ *不} 在 家 休息,

B: (Bù,) wǒ {méi/*bù} ZAI jiā xiūxi,

no 1SG NEG/ NEG PROGR+at home rest

我 在 整理 房间。

wǒ zài zhěnglǐ fángjiān.

1SG PROGR tidy room

‘(No,) I'm not having a rest at home,

I'm tidying up my room.’

(59) 他 {没 /*不} 在 床 上 躺 着。

Tā {méi/*bù} ZAI chuáng shàng tǎng-zhe.

3SG NEG/ NEG PROGR+at bed on lie -DUR

‘He's not lying on the bed.’

(Lin Jo-wang 2003: 431, [14b–c], glosses slightly changed)

The underlying presence of the progressive *zài* 在 requiring *méi* 没 is particularly clear in (59), because positional verbs such as *tǎng* ‘lie’ must be suffixed by *-zhe* 着 when selected as complement by *zài* 在 (cf. [47] above). (Note that this is not what Lin Jo-wang (2003) says, because he is not aware of the phenomenon of haplology). The question-answer pair in (58a–b) refers to an ongoing activity at speech time, hence with an underlying progressive *zài* 在 in (58a) and the first clause of (58b), and an overt instance of aspectual *zài* 在 in the second clause.

If we now compare these with sentences where the negation *bù* 不 precedes the VP with a *zài* 在-PP adjunct, we note that they are only acceptable under a generic, habitual reading, not an episodic one as in (58) and (59), as also reflected in the different choice of adverbs. There is thus no underlying progressive aspect, and *zài* 在 is a “pure” preposition here.

- (60) 他 {一般 /经常 /从来} 不 在家 学习/ 弹 琴。
 Tā {yībān / jīngcháng/cónglái} bù zài jiā xuéxí/ tán qín.
 3SG generally/often /ever NEG at home study/ pluck piano
 ‘He {generally/often} does not study/ play the piano at home.’
 ‘He never studies/ plays the piano at home.’

This ties in with the generalization above that activity verbs without any aspect marking are only acceptable in root sentences when giving rise to a habitual reading. Given the general consensus that habituais are semantically statives, the negation with *bù* 不 is expected, as also acknowledged by both Ernst (1995) and Lin Jo-wang (2003). This is further illustrated in (61a), this time without any *zài* 在 PP:

- (61) a. 她 不 抽 烟 /吃 鸡肉。
 Tā bù chōu yān /chī jīròu.
 3SG NEG inhale cigarette /eat chicken
 ‘She doesn’t smoke/eat chicken.’
 b. 她 没 抽 烟 /吃 鸡肉。
 Tā méi chōu yān /chī jīròu.
 3SG NEG inhale cigarette /eat chicken
 ‘She didn’t smoke/eat chicken.’
 (Lin Jo-wang 2003: 434, [17a–b], slightly changed)

When the negation *méi* 没 is used instead of *bù* 不, we obtain an episodic interpretation referring to a single event (cf. [61b]).

5.3 *Bù* 不 VP: Covert Presence of a Volitional Modality

Another case where *bù* 不 does not combine with a stative predicate, but with an activity verb in fact involves a covert “future or volition modality”, as per C.-T. James Huang’s (1988: 290) by now classical analysis of (62a–b) (his [7a], [45b]). (Also cf. Teng Shou-hsin 1974: 89.)

(62) a. 他们 不 跑 得 快。

Tāmen bù pǎo de kuài.

3PL NEG run DE fast

‘They won’t run fast.’

b. 他 不 跑 得 快，是 因为 他 不 想 跑。

Tā bù pǎo de kuài, shì yīnwèi tā bù xiǎng pǎo.

3PL NEG run DE fast be because 3SG NEG want run

‘He won’t run fast, because he doesn’t want to run.’

According to Huang (1988: 290), (62a) “cannot be understood as a general statement about them that they don’t run fast”. This is confirmed by the continuation in (62b), which clearly brings forward the covert presence of the future/volition modality in the first clause. Like modal auxiliaries in general, this covert auxiliary is likewise negated by *bù* 不.

This view has remained the consensus up to now and is adopted by Ernst (1995: 671) and Lin Jo-wang (2003: 429), who provides further examples for *bù* 不 negating the covert modal verb rather than the overt activity verb, thus giving rise to the meaning ‘do not want to V’, ‘won’t V’.²⁹

(63) a. 我 不 去 台北。

Wǒ bù qù Táiběi.

1SG NEG go Taipei

‘I don’t want to/will not go to Taipei.’

b. 他 不 帮 我 写 作业。

Tā bù bāng wǒ xiě zuòyè.

3SG NEG help 1SG write homework

‘He doesn’t want to help me with my homework.’

(Lin Jo-wang 2003: 429, [8a–b], glosses slightly changed)

These data tie in with Sun Hongyuan’s (2014: 218, [81a–c]) observation that for [+intentional] activity verbs describing eventualities that can be scheduled, controlled by an agent, a futurate interpretation can be obtained *without* the otherwise obligatory future auxiliary *huì* 会:

(64) a. 中国 队 明天 比赛。

Zhōngguó duì míngtiān bǐsài.

China team tomorrow play

‘The Chinese team plays tomorrow.’

- b. 中国 队 明天 *(会) 赢。
 Zhōngguó duì míngtiān *(huì) yíng.
 China team tomorrow will win
 ‘The Chinese team will win/*wins tomorrow.’

The same empty volitional/future modality as postulated by Huang (1988) for *bù* 不 preceding activity verbs in a non-habitual reading can be assumed here. The more so as the verbs in the examples used to illustrate this phenomenon all seem to involve control by the agent, although this observation needs to be confirmed by further research.

5.4 ‘*Bù* 不 ‘not’ (*shì* 是 ‘be’) *zài* 在 VP’ in Correcting Statements and Questions

We have seen in (58a) (repeated in [65] below) that the progressive aspect *zài* 在 requires the negation *méi* 没, including the cases where it is only underlyingly present, because phonologically fused with the following preposition *zài* 在:

- (65) a. A: 你 在 家 休息 吗?
 A: Nǐ ZÀI jiā xiūxi ma?
 2SG PROGR+at home rest SFP
 ‘Are you having a rest at home?’
 b. B: (不,) 我 没 /*不 在 家 休息,
 B: (Bù,) wǒ méi/*bù ZÀI jiā xiūxi,
 no 1SG NEG/ NEG PROGR+at home rest
 我 在 整理 房间。
 wǒ zài zhěnglǐ fángjiān.
 1SG PROGR tidy room
 ‘(No,) I’m not having a rest at home,
 I’m tidying up my room.’

While (65b) is a well-formed answer to (65a) and negation with *bù* 不 is clearly unacceptable, another, slightly more natural answer to (65a) features negation with *bù shì* 不是 ‘not be’:

- (66) (不,) 我 不 是 在 家 休息,
 (Bù,) wǒ bù shì ZÀI jiā xiūxi,
 no 1SG NEG be PROGR+at home rest
 我 在 整理 房间。

wǒ zài zhěnglǐ fángjiān.

1SG PROGR tidy room

‘(No,) I’m not having a rest at home, I’m tidying up the room.’

As is well-known, *bù shì* 不是 ‘not be’ has scope over the sequence to its right and accordingly can either negate the entire sequence (cf. [67a]) or a subpart of it (cf. [67b]):

- (67) a. (不,) 我 不 是 [_{VP} 在 家 休息],
 (Bù,) wǒ bù shì [_{VP} ZÀI jiā xiūxi],
 no 1SG NEG be PROGR+at home rest
 我 在 公园 里 跑 步。
 wǒ ZÀI gōngyuán lǐ pǎo bù.
 1SG PROGR+at park in run step
 ‘(No,) it’s not the case that I’m having a rest at home,
 [but] I’m running in the park.’
- b. (不,) 我 不 是 [_{VP} 在 家 休息],
 (Bù,) wǒ bù shì [_{VP} ZÀI jiā xiūxi]
 no 1SG NEG be PROGR+at home rest
 我 是 在 公园 里 休息。
 wǒ shì ZÀI gōngyuán lǐ xiūxi.
 1SG be PROGR+at park in rest
 ‘(No,) it’s not the case that I’m having a rest at home,
 I’m having a rest in the park.’

In the propositional negation with *bù shì* 不是 ‘not be’, *shì* 是 ‘be’ takes as complement the extended verbal projection to its right, as indicated in the bracketing. When the copula is negated, negation can affect the complement as a whole or a subpart, as in the case of English ‘it is not (the case) that’. (Cf. Paul and Whitman 2008 for discussion and references.)

Negation with *bù shì* 不是 ‘not be’ = ‘it is not the case that’ is either used in declaratives to correct a wrong statement (cf. [66–67] above), or in questions asking for a confirmation of a prior statement or an assumption held by the speaker:

- (68) 你 不 是 在 写 作业 吗? 怎么 出来 了?
 Nǐ bù shì zài xiě zuòyè ma? Zěnmē chū-lái -le?
 2SG NEG be PROGR write homework SFP how exit-come-PERF
 ‘Aren’t/weren’t you (supposed to be) doing your homework?
 How come you have gone out?’

These two contexts for *bù shì* 不是 ‘not be’ are important in order to come to terms with sequences of the form ‘*bù* 不 *zài* 在 VP’ and ‘*bù* 不 *zhèng* 正 *zài* 在 VP’ present in online corpora, which at first sight seem to contradict the robust generalization that the progressive aspect must be negated by *méi* 没, not by *bù* 不. However, a covert *shì* 是 ‘be’ is present here.³⁰

- (69) a. 你们 为什么 现在 不 [是] 在 干 活?
 Nimen wèishénme xiànzài bù [shì] zài gàn huó?
 2PL why now NEG be PROGR work life
 ‘Why is it that you are not working right now?’
 ‘How come you are not working right now?’
 b. 我 不 [是] 正 在 上 课 吗?
 Wǒ bù [shì] zhèng zài shàng kè ma?
 1SG NEG be just PROGR attend class SFP
 ‘Isn’t it the case that I’m attending classes right now?’
 ‘Am I not attending classes right now?’

The native speakers consulted agreed that the interrogative nature and the linguist context show (69a–b) to be (rhetorical) questions with *bù shì* 不是 ‘not be’. This is confirmed by the fact that (70) is only acceptable when followed by the second clause, which indicates its interrogative nature:

- (70) 你们 现在 不 [是] 在 干 活, *(在 干 什么)?
 Nimen xiànzài bù [shì] zài gàn huó, *(zài gàn shénme)?
 2SG now NEG be PROGR do life PROGR do what
 ‘If it’s not the case that you’re working right now, then what are you doing?’

Here again, *bù* 不 is to be analysed as *bù shì* 不是 ‘not be’ with the associated semantics. If one wanted to negate the activity itself, *méi* 没 would have to be used instead of *bù* 不, resulting in ‘You are not working now.’ for the first clause.

Finally, *bù* 不 functioning as propositional negation *bù shì* 不是 ‘not be’ = ‘it is not the case that’ in questions asking for confirmation or in declaratives correcting a prior statement can likewise nicely account for the observations by reviewer 1 concerning the acceptability of *bù* 不 (in addition to *méi* 没) in (49b) and (59), repeated in (71) and (72) below:

- (71) 他 也许 /大概 没 /不 在 上 课。
 Tā yěxǔ /dàgài méi /bù zài shàng kè.
 3SG perhaps/probably NEG /NEG PROGR attend class
 ‘He is perhaps/probably not attending classes.’
- (72) 他 没 /不 在 床 上 躺 着。
 Tā méi /bù ZÀI chuáng shàng tǎng-zhe.
 3SG NEG/NEG PROGR+at bed on lie -DUR
 ‘He’s not lying on the bed.’

Concerning (72), reviewer 1 further notes that *bù* 不 is fine when (72) answers the question in (73):

- (73) 他 在 床 上 躺 着 吗?
 Tā ZÀI chuáng shàng tǎng-zhe ma?
 3SG PROGR+at bed on lie -DUR SFP
 ‘Is he lying on the bed?’
- (74) a. 不, 他 不 是 在 床 上 躺 着。
 Bù, tā bù shì ZÀI chuáng shàng tǎng-zhe.
 no 3SG NEG be PROGR+at bed on lie -DUR
- b. 他 (是) 在 地 上 躺 着。
 Tā (shì) ZÀI dì shàng tǎng-zhe.
 3SG be PROGR+at floor on lie -DUR
 ‘No, it’s not the case that he is lying on the bed.
 He’s lying on the floor.’

As evident from the continuing clause in (74b) (where *shì* 是 can be added), 不 *bù* in (72) is a clear case of *bù(shì)* 不(是) ‘not (be)’ (cf. [74a] correcting the assumption made in the preceding question [73]).

Similarly, *bù* 不 in (71) (repeated in [76a]) is acceptable because functioning as propositional negation *bù shì* 不是 ‘not be’, with (75) provided by us as a possible context:

- (75) 他 这 个 时 间 在 上 课 吗?
 Tā zhè gè shíjiān zài shàng kè ma?
 3SG this CL time PROGR attend class SFP
 ‘Is he attending classes at this moment?’
- (76) a. (这 个 时 间) 他 也 许 /大 概
 (Zhè gè shíjiān) tā yěxǔ /dàgài
 this CL time 3SG perhaps/ probably
 不 (是) 在 上 课。

- bú (shì) zài shàng kè.
 NEG be PROGR attend class
 b. 可能 (是) 在 辅导 学生。
 Kěnéng (shì) zài fǔdǎo xuéshēng.
 perhaps be PROGR supervising student
 ‘He is perhaps/probably not attending classes (at this moment).
 He is probably supervising students.’

The continuation (76b) shows that the entire VP ‘attend class’ is negated here and replaced by the VP ‘supervise students’ as the correct event.

5.5 Wrap-up

The AspP headed by the progressive *zài* 在 must be negated by *méi* 没, not by *bù* 不. ‘*zài* 在 VP’ thus behaves on a par with derived states of the form ‘V-*zhe* 着’ which likewise require the negation *méi* 没. The fact that some stative verbs as well as a subset of modal auxiliaries allow *méi* 没 in addition to *bù* 不 further confirms the compatibility of *méi* 没 with states and directly challenges Lin Jo-wang’s (2003) scenario with *méi* 没 as exclusive negation for activities.

Furthermore, the choice of negation provides additional evidence for the phenomenon of haplogy: ‘*méi* 没 [*zài* 在 NP] VP’ indicates the underlying presence of the progressive aspect *zài* 在, whereas ‘*bù* 不 [*zài* 在 NP] VP’ either indicates a generic statement or an interpretation with an underlying volitional/future modality: ‘not want to/will not VP [at NP]’.

6. CONCLUSION

Having removed the cumbersome preliminary obstacle of the status of *zhèng* 正 *zài* 在, which is not one word, but the adverb *zhèng* 正 ‘just, right then’ preceding the progressive aspect head *zài* 在, we addressed some of the central issues concerning the progressive aspect *zài* 在. (For the small emerging group of 1-word *zhèngzài* 正在 speakers, cf. the appendix.)

We argued in favour of the constituency [_{AspP} *zài* 在 [V-*zhe* 着]] and demonstrated important syntactic and semantic differences between the progressive aspect *zài* 在 and the durative aspect -*zhe* 着, often blurred by their co-occurrence in ‘*zài* 在 V-*zhe* 着’ and their selecting nearly identical verb classes. Importantly, unlike ‘*zài* 在 V’, ‘V-*zhe* 着’ is only acceptable as the sole predicate in a root sentence when it can be interpreted as a

derived state. Otherwise, the presence of the progressive aspect *zài* 在 is required: ‘*zài* 在 V-*zhe* 着’.

This crucial role of *zài* 在 is often obscured by its phonological fusion with the preposition *zài* 在 ‘at’ heading an adjacent PP (*haplology*):

(77) NP [_{Asp} *zài* 在 [_{VP} [_{PP} *zài* 在 NP] [V-*zhe* 着 NP]]]
 => NP *ZÀI* 在 NP V-*zhe* 着 NP

The presence/absence of an underlying progressive aspect *zài* 在 (having fused with the preposition *zài* 在) is reflected in the choice of negation. Since the progressive aspect *zài* 在 must be negated by *méi* 没, not *bù* 不, it follows that no underlying aspectual *zài* 在 is present when *bù* 不 negates a VP with a *zài* 在 PP adjunct: NP *bù* 不 [_{VP} [_{PP} *zài* 在 NP] VP]. When there is an underlying aspectual *zài* 在, the negation *méi* 没 is required, resulting in the surface sequence ‘NP *méi* 没 [*ZÀI* 在 NP VP]’. Finally, given that a subset of auxiliaries, some stative verbs and state-denoting ‘V-*zhe* 着’ are negated by *méi* 没, negation of ‘*zài* 在 VP’ by *méi* 没 is compatible with the status of ‘*zài* 在 VP’ as a state, hence able to be true at a moment. The exact nature of these different states must be left for further research.

NOTES

1. The following abbreviations are used in glossing examples: CL classifier; DUR durative aspect (-*zhe*); EXP experiential aspect (-*guo*); NEG negation; PERF perfective aspect (-*le*); PL plural; PROGR progressive aspect (*zài*); SFP sentence-final particle; SG singular; SUB subordinator (*de*).

2. Given the controversial syntactic status of *de* 得, we follow Huang (1988: 275) who simply glosses it as *DE*. More than thirty years later, not much progress has been made. But cf. Paul (2015, ch. 8.3.3; 2017) for the analysis of *de* 得 as a functional head selecting predicative adjectives, with the resulting DeP (a secondary predicate) subsequently merging with V°.

3. Similarly, Lin Jo-wang (2002: 268) assigns adverb status to *zhèng* 正 ‘just’, but treats it on a par with the progressive aspect *zài* 在, *modulo* the “additional tint of more emphasis on the reference time at which the

proposition is asserted to be true” for *zhèng* 正. By contrast, Lin Jo-wang (2017) treats *zhèngzài* 正在 ‘right at’ as an “aspectual adverb”, i.e., as one word.

4. For the ‘1-word *zhèngzài* 正在’ speakers to be presented in the appendix, *zhèngzài* 正在 is never the aspectual head, i.e., a bisyllabic variant of the aspectual *zài* 在, either, but clearly an adverb ‘just, right then’, as evidenced by the well-formedness of [_{adv} *zhèngzài* 正在] [_{Asp°} *zài* 在] VP’.

5. Manner adverbs form a subset within the VP-level adverbs, for they must follow auxiliaries (cf. Paul 2017: 123, [2b] and references therein).

6. The presence of *zhèng* 正 ‘just, right then’ in (12a) is acceptable for some speakers, provided the sentence is continued as in (i):

- (i) 司机 还 正 在 唠叨, 他 就 下 车 了。
 Sījī hái zhèng zài lāodao, tā jiù xià chē le.
 driver still just PROGR chatter 3SG then descend car SFP
 ‘The driver was still chattering, when he got out of the car.’

7. The additional evidence against *zhèng* 正 *zài* 在 as one word provided by reviewer 2 is gratefully acknowledged. It would, however, be insufficient on its own, because it could always be argued that the sequence ‘*zhèng* 正 *zài* 在 V’ can be analysed in two ways, as one word when contiguous and as two words when separated by a manner adverb as in (13a). This is also the reason why the emerging 1-word *zhèngzài* 正在 group and the consequences of wordhood for *zhèngzài* 正在 discussed in the appendix are important: given that *zhèngzài* 正在 is an adverb ‘just, right then’, the progressive marker *zài* 在 is added when required: [_{Adv} *zhèngzài* 正在] [_{Aspect} *zài* 在 VP].

8. Tsai Wei-Tien (2008: 678, 684) provides no evidence for the alleged raising of the aspectual head *zài* 在 to T° and thus into the same projection (TP) that hosts the subject. His only example with *zài* 在 does not feature any adverb and can therefore not tell us anything about *zài* 在’s hierarchical position (in or above vP) (cf. Pollock (1989) for adverbs as a diagnostic for V-movement):

- (i) 阿 Q 在 哭。
 Akiu zài kū.
 Akiu PROGR cry
 ‘Akiu is crying.’

(Tsai Wei-Tien 2008: 684, [4a])

9. Li Tianshu (2015: 12, 49) transposes into *syntax* Lin Jo-wang's (2002: 272) stipulation that *-zhe* 着 takes scope over *zài* 在 in the corresponding "logical structure": [*zhe* 着 [*zài* 在 [subject NP V object NP]]]. Lin (2002) does not comment on the precise status of the postulated "logical structure" nor on how it is to be mapped into syntax, where *zài* 在 must occur below the subject and where *-zhe* 着 is a verbal suffix: (**zài* 在) [_{TP} subject NP [_{AspP} *zài* 在 [_{VP} V *-zhe* 着 NP (**-zhe* 着)]]]. In addition, Lin (2002) cannot correctly predict the data in (22) below where *-zhe* 着 first combines with the verb and where the resulting 'V-*zhe* 着' is then selected as complement by *zài* 在, leading to a hierarchical relation which is the exact opposite of his logical structure.

10. The unacceptability of **zài* 在 *zuò* 坐 'PROGR sit' was already noticed by Teng Shou-hsin (1975: 129) and Chen Chung-yu (1978: 91). Interestingly, positional statives such as *sit*, *stand* and *lie* form an exceptional class in English as well, insofar as they can occur in the progressive with a *non*-dynamic interpretation (cf. Ramchand 2018: 44, 215, note 3).

11. Chen Gang (1980) and Kong Lingda (1994) characterize sentences such as (23a–b) as "incomplete", a term also used by a.o. Tang and Lee (2000) and Tsai Wei-tian (2008). However, the latter do not take into account the well-documented fact that 'V-*zhe* 着' is a licit predicate when referring to a derived state (cf. [25a–c] below). Nor do they comment on the well-known root vs non-root asymmetry. Cheng Gang (1980: 23–24) e.g., contrasts the constrained acceptability of 'V-*zhe* 着' in main clauses with its unconstrained acceptability in non-main clauses, where it can either denote an activity or a derived state.

12. The contrast between (23b) and (24b) straightforwardly invalidates Lin Jo-wang's (2002: 268) proposal to treat the allegedly rarely discussed adverb *zhèng* 正 'just' on a par with the progressive aspect marker *zài* 在.

13. The acceptability of 'V-*zhe* 着' in imperatives is a different case, a fact reviewer 1 reminds us of (cf. Lu Jianming 1999: 331–332):

- (i) 蹲 着, 别 起来!
 Dūn -zhe, bié qǐlái!
 squat-ZHE NEG rise
 'Remain squatted, don't get up!'

14. Another way to make the sentence finite, hence acceptable as a root sentence, is to add the *sentence-final particle* (SFP) *ne* 呢, as shown by Kong

Lingda (1994: 440) for (23a–b), illustrated here only for (23b), repeated as (i) (cf. Chen Gang 1980: 23 for the same observation):

- (i) 他 正 写 着 信 *(呢)。
 Tā zhèng xiě -zhe xìn *(ne).
 3SG just write-DUR letter SFP
 ‘He’s just writing a letter.’

Ne 呢 can help to anchor an unbounded activity and hence make the sentence finite, because it provides a contextual interval for *V-zhe* 着 which stretches over time itself (cf. Ramchand 2020). This also holds for (31b) below, which becomes acceptable as a root sentence when *ne* 呢 is added (cf. [ii]):

- (ii) 汽车 飞奔 着 *(呢)。
 Qìchē fēibēn-zhe *(ne).
 car gallop-DUR SFP
 ‘Cars are galloping.’

For the role of the low SFPs (*ne* 呢, *le* 了, *lái* 来着) in the split CP with respect to finiteness, cf. Paul (2018), Ramchand (2020), Paul and Ramchand (2023). For the overall structure of the Chinese split CP, cf. a.o. Paul (2014), Pan (2019), Paul and Yan (2022) and references therein.

15. For states (i.e., stative verbs and adjectives), an appropriate temporal adverb is sufficient to indicate past tense (cf. [ia], [iia]), whereas an auxiliary verb is required to indicate future tense, as in (ib) and (iib) (cf. examples below from Sun Hongyuan 2014: 44–45, 61, 165):

- (i) a. (那 时候) 老谋子 喜欢 巩俐。
 (Nèi shíhòu) Lǎomóuzi xǐhuān Gǒnglì.
 that time Laomouzi love Gongli
 ‘Lǎomóuzi loves Gǒnglì.’ / ‘At that time, Lǎomóuzi loved Gǒnglì.’
 b. 老谋子 有一天 *(将会) 喜欢 巩俐 吗?
 Lǎomóuzi yǒuyītiān *(jiānghuì) xǐhuān Gǒnglì ma?
 Laomouzi one.day FUT love Gongli SFP
 ‘Will Laomouzi one day love Gongli?’

- (ii) a. (昨天) 璐璐 很 沮丧。
 (Zuótiān) Lùlù hěn jǔsàng.
 yesterday Lulu very be.frustrated
 ‘Lulu is frustrated.’/ ‘Lulu was frustrated yesterday.’
- b. 明天 璐璐 *(将会) 很 沮丧。
 Míngtiān Lùlù *(jiānghuì) hěn jǔsàng.
 tomorrow Lulu will very be.frustrated
 ‘Lulu will be frustrated tomorrow.’

16. The root vs non-root dichotomy for ‘V-*zhe* 着’ is very sharp for Northern Mandarin speakers, while this is not always the case for Taiwanese Mandarin speakers. Lin Jo-wang seems to be on a par with Northern speakers, because elsewhere he notes the unacceptability of ‘V-*zhe* 着’ as root predicate:

- (i) *张三 盖 着 一 栋 房子。
 *Zhāngsān gài -zhe yī dòng fángzi.
 Zhangsan build-DUR 1 CL house

(Lin Jo-wang 2002: 260, [5])

The present study is based on the judgements by Northern Mandarin speakers.

17. Some speakers prefer (32a,c) and (33a,c) without *-zhe* 着 and require emphasis on the PP in the (a) cases. By contrast, *-zhe* 着 must be present in (31c), which is understood as describing a scene.

18. In a habitual interpretation, *zài* 在 in ‘NP *zài* 在 NP VP’ realizes the preposition only, as confirmed by the optionality of the *zài* 在 PP:

- (i) 他 平常 (在 家里) 睡 午觉。
 Tā píngcháng (zài jiālǐ) shuì wǔjiào.
 3SG in.general at home sleep nap
 ‘In general he takes a nap (at home).’

19. According to Chen Chung-yu (1978: 79), native speakers produce (34c), not (34b), when asked to “subtract” the location information from (34a).

20. This is not what Chen Chung-yu (1978: 91) says herself. Instead, she uses (35) to illustrate the special semantic properties of positional verbs.

21. Haplogy of the perfective aspect suffix *-le* 了 and the SFP *le* 了 when the verb occurs in sentence-final position also requires adjacency (cf. a.o. Chao Yuen Ren 1968: 247): [CP [TP NP V-le 了] le 了] => ‘NP V LE 了’.

22. Sentences of the type illustrated in (40) – (41) are only acceptable without the aspectual *zài* 在 in TP when they allow a generic, habitual reading (also cf. Sun Hongyuan 2014: 92–98):

- (i) [TopP [PP 在 里屋] [TP 他 每 天 都 看 书]].
 [TopP [PP Zài lǐwū] [TP tā měi tiān dōu kàn shū]].
 at room 3SG every day all see book
 ‘He reads in the backroom every day.’

23. This is not what Chen Chung-yu (1978: 83) says. For her, ‘*zài* 在 V-*zhe* 着’ is a case of emphasis on the progress of an action.

24. Ernst (1995: 694) incorrectly claims *bù* 不 to be the correct negation of ‘*zài* 在 VP’ for Southern speakers, while *méi* 没 must be used by Northern speakers. This is invalidated by Lin (2003: 454, note 3), stating that only ‘*méi* 没 *zài* 在 VP’ is correct for all speakers. Also cf. Hsieh Miao-Ling (2001: 23).

25. For the reason why *bù* 不 is likewise acceptable for reviewer 1 here, cf. the end of section 5.4 below.

26. The auxiliaries *huì* 会 ‘will’, *gāi* 该 ‘should’, *yīnggāi* 应该 ‘should, must’, *kěyǐ* 可以 ‘can, may’, *yuànyì* 愿意 ‘wish’ are only compatible with *bù* 不 (cf. Lü Shuxiang 2000: 384).

27. Lin Jo-wang’s (2003: 431, [12a–b]) own example is unacceptable, due to the presence of the quantified NP ‘one picture’ in the scope of negation:

- (i) *墙 上 没/ 不 挂 着 [一 幅 画].
 *Qíáng shàng méi/ bù guà -zhe [yī fú huà].
 wall on NEG/ NEG hang-DUR 1 CL picture
 ‘There is not a picture hanging on the wall.’
 (Lin Jo-wang 2003: 431, [12a–b]; glosses slightly changed)

29. This is different from “derived states” in the sense of Lin Jo-wang (2003: 432) who uses this term for the predicate in the locative construction (55), where the intended state is said to require an initial input of energy.

30. *Bù* 不 can only negate ‘V-*zhe* 着’ under a volitional reading, i.e., in the covert presence of a modal (cf. Lin Jo-wang 2003: 455, note 4, [i]). Hsieh Miao-Ling (2001: 40, [35a–b]) likewise contrasts the volitional reading in (ia) with the negation of the episodic reading in (iib):

- (i) 你 不 在 床 上 躺 着, 你 要 去 哪 里?
 Nǐ bù zài chuáng shàng tǎng-zhe, nǐ yào qù nǎlǐ?
 2SG NEG at bed on lie -DUR 2SG want go where
 ‘You don’t want to lie on the bed. Where do you want to go?’
- (ii) a. 他 不 坐 着 / 站 着。
 Tā bù zuò-zhe/ zhàn -zhe.
 3SG NEG sit-DUR / stand-DUR
 ‘He does not want to be sitting/standing.’
 b. 他 没 坐 着 / 站 着。
 Tā méi zuò-zhe/ zhàn -zhe.
 3SG NEG sit-DUR / stand-DUR
 ‘He is not sitting/standing.’

31. The examples in the BLU-BCC corpus (<http://bcc.blcu.edu.cn>) are all (rhetorical) questions with the copula dropped: NP *bù* 不 ~~shì~~ 是-*zhèng* 正 *zài* 在 VP? ‘NP NEG be just PROGR VP?’ = ‘Isn’t it the case that NP is just V-ing?’.

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汉语进行体标记“在”新议

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摘要

本文探讨迄今为止在汉语进行体标记“在”及相关研究方面被忽略的问题。我们赞成进行体“在”和持续体“着”共现的句法结构为 [AspP 在 [V-着]]，并认为其体现了二者之间重要的句法和语义区别。不同于“在 V”，当“V-着”表示派生的同质性状态时，它只在根句作为主要谓词才会被接受。否则，进行体“在”需要出现。“在”的这一关键性作用常常由于引导介词短语的与它语音融合的介词“在”的出现而被掩盖。深层结构的进行体“在”的隐现反映在否定词的选择上。由于进行体“在”必须被“没”否定，而非“不”，所以当“不”否定包含“在 PP”附加语的 VP 时，进行体“在”不出现。相反，当进行体“在”出现时，句子被否定时要求“没”出现，从而出现表层结构的顺序为“NP 没 [[在 NP] VP]”。

关键词

持续体 -着 合体 (haplology) 否定词“不”和“没” 限定性

APPENDIX

As mentioned in the introduction, there are a handful of examples attested in corpora where *zhèngzài* 正在 ‘just, right then’ is an adverb, i.e., one word, as evidenced by its co-occurrence with the aspectual *zài* 在, leading to the sequence: [_{AspP} [_{adv} *zhèngzài* 正在] [_{AspP} *zài* 在 VP]] (cf. [2a-d] below). More precisely, concerning the two corpora consulted (in November 2021), the PKU-CCL corpus (http://ccl.pku.edu.cn:8080/ccl_corpus) features ten examples and the BLU-BCC corpus (<http://bcc.blcu.edu.cn>) twenty-four examples. Three different structures can be found.

i) ‘*zhèngzài* 正在 [_{PP} *zài* 在 NP] VP’

It makes up for half of the cases (6 of 10 examples in the PKU-CCL corpus, and 12 of 24 in the BLU BCC corpus, respectively). The majority involves haplology (noted as *zài* 在) of the progressive aspect *zài* 在 and the preposition *zài* 在 of the immediately adjacent *zài* 在-PP. The progressive aspect *zài* 在 must be present underlyingly to explain the non-habitual reading of what at the surface looks like bare verbs. (Recall from section 4 that non-stative bare verbs are only acceptable as sole predicates in a root sentence when giving rise to a habitual reading.) This analysis is confirmed by sentence (1d) where the adjunct PP is headed by *hé* 和 ‘with’ and preceded by both the adverb *zhèngzài* 正在 ‘just’ and the aspectual *zài* 在:

(1a) 这群人正在在我家打边炉。

Zhè qún rén zhèngzài zài wǒ jiā dǎ biānlú.
this group people just PROG+at 1sg home strike side.stove
‘This group of people is just enjoying a Cantonese hotpot
at my place.’

(1b) 比赛正在在兰州举行。

Bǐsài zhèngzài zài Lánzhōu jǔxíng.
competition just PROG+at Lanzhou hold
‘The match is just going on in Lanzhou.’

(1c) 这部电视剧正在在北京拍摄。
 Zhè bù diànshìjù zhèngzài zài Běijīng pāishè.
 this cl TV.drama just PROGR+at Beijing shoot
 ‘This TV drama is just being filmed in Beijing.’

(1d) Tā [AspP 正在 [AspP 在 [PP 和朋友] 交谈]].
 Tā [AspP zhèngzài [AspP zài [PP hé péngyǒu] jiāotán]].
 3sg just PROGR with friend chat
 ‘He is just chatting with friends.’

ii) ‘zhèngzài 正在 [AspP zài 在 VP]’

The co-occurrence of *zhèngzài* 正在 with the aspectual *zài* 在 immediately preceding a VP clearly points to the adverbial status of *zhèngzài* 正在 ‘just, right then’.

(2a) 他正在在学习中文。
 Tā zhèngzài zài xuéxí zhōngwén.
 3sg just PROGR learn Chinese
 ‘He is just learning Chinese.’

(2b) 同事们正在在购买礼品。
 Tóngshìmen zhèngzài zài gòumǎi lǐpǐn.
 colleagues just PROGR buy gift
 ‘The colleagues are just buying gifts.’

(2c) 积雪正在在迅速地旋转着。
 Jīxuě zhèngzài zài xùnsùde xuánzhuǎn-zhe.
 snow just PROGR rapidly spin -dur
 ‘The snow is just spinning rapidly.’

(2d) 渔业资源正在在急剧减少。
 Yúyè zīyuán zhèngzài zài jíjù jiǎnshǎo.
 fishery resource just PROGR sharp decrease
 ‘Fishery resources are just declining sharply.’

iii) ‘zhèngzài 正在 [VP [V° zài 在] PostP]’

In this structure, the adverb *zhèngzài* 正在 ‘just’ precedes the (stative) verb *zài* 在 ‘be at’, which in turn selects a postpositional phrase as its complement. Given the well-known fact that the verb *zài* 在 ‘be at’ is incompatible with any aspect marker, this again confirms the status of *zhèngzài* 正在 as an adverb, on a par with *zhèng* 正 ‘just’, not with the aspectual head *zài* 在.

(3a) 各方保证[VP 正在 [VP 在[PostP 会谈中]]]。

Gè fāng bǎozhèng [VP zhèngzài [VP zài [PostP huìtán zhōng]]]
each side assure just be.at talks in
‘All parties assure to be presently (engaged) in talks.’

(3b) 他似乎此刻正在在冷风里。

Tā sìhū cǐkè zhèngzài zài lěng fēng lǐ.
3sg seemingly at.present just be.at cold wind in
‘It seems as though he is just in a cold wind right now.’

(3c) 本公司的“蓝星大厦”正在在建设中。

Běn gōngsī de ‘lán -xīng-dàshà’ zhèngzài zài jiànshè zhōng.
IPL company SUB blue-star -building just be.at build in
The “blue star” building of our company is just under construction.’

(3d) 电话正在在接通中。

Diànhuà zhèngzài zài jiētōng zhōng.
phone just be.at put.through in
‘The phone is just connecting.’

While the majority of the 1-word-*zhèngzài* 正在 examples are from the microblogging website *Weibo* 微博 and hence probably rather recent (although no year is indicated here), there are also some dated occurrences from other text genres.

The earliest attestation for ‘*zhèngzài* 正在 [PP zài 在 NP] VP’ (group 1) is 1979 (in the Chinese translation of Rachel Carson’s *Silent spring* from 1962). For group 2, ‘*zhèngzài* 正在 [AspP zài 在 VP]’, we have 1998 (in Ren Yihong’s *Ài nǐ bù liú yú dì* 爱你不留余地). Concerning group 3,

zhèngzài 正在 [_{VP} [_{V°} *zài* 在] PostP]], it is first attested in a *Rénmín Rìbào* 人民日报(People's Daily) article from 1997. This is noteworthy because “negligence” as a consequence of the fast writing mode in *Weibo* 微博 can be excluded as a possible factor for the use of the adverb *zhèngzài* 正在 ‘just, right then’. More importantly, it nicely illustrates the difference between a change itself, more precisely the creation of a new adverb *zhèngzài* 正在 ‘just, right then’ and its *diffusion* among the speakers evolving in time, with an increase in the period covered by *Weibo* 微博. (Cf. Hale 2007 for extensive discussion of *change* vs *diffusion* of that change.)

The examples in (1) – (3) above (occasionally shortened in parts not relevant for the problem under discussion) also provided the material for our informal online questionnaire, whose aim was to find speakers who actually use the adverb *zhèngzài* 正在 in combination with the aspectual *zài* 在 +VP. All the sentences were double-checked for their acceptability with *zhèng* 正 ‘just’ instead of *zhèngzài* 正在 ‘just’ for the “canonical” group of native speakers, as represented by one of the co-authors. In addition, we chose two [+human] subjects and two [-human] subjects for each group, even though we think it highly unlikely that this could have any influence on the outcome.

Sixty-one participants took part in the online questionnaire where the twelve sentences above were augmented by twelve filler sentences and presented in a randomized order. Five judgement types were offered: completely acceptable (100%), probably acceptable (??OK), probably unacceptable (??*), completely unacceptable (*) and “I don’t know” (Ø). While detailed information on the judgements per sentence is provided in the overview at the end of the appendix, the results can be summarized as follows.

The structure in (1) ‘*zhèngzài* 正在 [_{PP} *zài* 在 NP] VP’ with a preverbal *zài* 在-PP, and the structure ‘*zhèngzài* 正在 [_{VP} [_{V°} *zài* 在] PostP]’ in (3) where the adverb *zhèngzài* 正在 ‘just’ precedes the verb *zài* 在 ‘be at’, received eleven and thirteen “100% acceptable” judgements, respectively; the structure in (2) ‘*zhèngzài* 正在 [_{AspP} *zài* 在 VP] with *zhèngzài* 正在 preceding ‘aspectual *zài* 在 + VP’ was judged “100% acceptable” only four times.

While the comparatively high frequency (within the modest sample at hand) of the adverb *zhèngzài* 正在 ‘just’ preceding a *zài* 在-PP looks plausible, because it can be linked to the phenomenon of haplology (cf. section 4 above), the comparatively high frequency of *zhèngzài* 正在 ‘just’ with the verb *zài* 在 ‘be at’ is at first sight somewhat surprising, given that the verb *zài* 在 is incompatible with any aspect marker and accordingly, there does not exist any potential source structure for the possible reanalysis of the adverb *zhèng* 正 ‘just’ and the aspectual *zài* 在 into the adverb *zhèngzài* 正在: *[_{Adv} *zhèng*] [_{AspP} *zài* 在 [_{V°} *zài* 在]]. The only possible parallel we can see is that with the adjunct NP *xiànzài* 现在 ‘now’, acceptable as modifier of the verb *zài* 在 ‘be at’, as in e.g., *Tā xiànzài zài jiā* 他现在在家 ‘3SG now be.at home’ = ‘She is at home now.’

There were only two speakers among the sixty-one who accepted all of the test sentences with either “100% acceptable” or “probably acceptable”. Our initial purpose of the questionnaire (besides confirming the findings in the corpora), i.e., being able to ask follow-up questions in order to further test the distribution of the adverb *zhèngzài* 正在 ‘just’ was therefore seriously hampered. We nevertheless checked two data points: the acceptability of *zhèngzài* 正在 ‘just’ (cf. 4a) in contexts where the canonical speakers only use the adverb *zhèng* 正 ‘just’ and where aspectual *zài* 在 is excluded (cf. [4b] and the discussion in section 2 above), and the acceptability of *zhèngzài* 正在 ‘just’ in the topic position before the subject (cf. [5a]), a position excluded for low VP adverbs such as *zhèng* 正 ‘just’ (cf. [5b]), but acceptable for adjunct NPs such as *xiànzài* 现在 ‘now’ (cf. [5c]):

- (4a) 你来得真巧，我[_{Adv} 正在] 要找你。
 Nǐ lái DE zhēn qiǎo , wǒ [_{Adv} zhèngzài] yào zhǎo nǐ.
 2SG come DE really opportune 1SG just want search 2SG
 (Intended: ‘You’ve come at the right time, just as I wanted to go see you.’)
- (4b) 你来得真巧，我 [_{Adv} 正] (*[_{Asp°} 在]) 要找你。
 Nǐ lái DE zhēn qiǎo , wǒ [_{Adv} zhèng] (*[_{Asp°} zài]) yào zhǎo nǐ.
 2SG come DE really opp. 1SG just PROGR want search 2SG
 ‘You’ve come at the right time, just as I wanted to go see you.’

- (5a) 放心吧, [Adv 正在] 呢,
 Fàng xīn ba, [Adv zhèngzài] ne,
 put heart SFP just TOP
 电话在 [PostP 接通中], 你准备说吧。
 diànhuà zài [PostP jiētōng zhōng], nǐ zhǔnbèi shuō ba.
 phone be.at put.through in 2SG prepare talk SFP
 (Intended: ‘Be reassured, the phone is just connecting, get ready to talk.’)
- (5b) 放心吧, (*正呢,)
 Fàng xīn ba, (*zhèng ne,)
 put heart SFP just TOP
 电话正在 [PostP 接通中], 你准备说吧。
 diànhuà zhèng zài [PostP jiētōng zhōng], nǐ zhǔnbèi shuō ba.
 phone just be.at put.through in 2SG prepare talk SFP
 ‘Be reassured, the phone is just connecting, get ready to talk.’
- (5c) 放心吧, 现在呢,
 Fàng xīn ba, xiànzài ne,
 put heart SFP now TOP
 电话(正)在 [PostP 接通中], 你准备说吧。
 diànhuà (zhèng) zài [PostP jiētōng zhōng], nǐ zhǔnbèi shuō ba.
 phone just be.at put.through in 2SG prepare talk SFP
 ‘Be reassured, the phone is just connecting right now, get ready to talk.’

However, the results of our mini-test were inconclusive; the sentences were either rejected as unacceptable or as too difficult to judge.

Data from the Acceptability Judgment Test

A. Participants' information (general): Total N=61

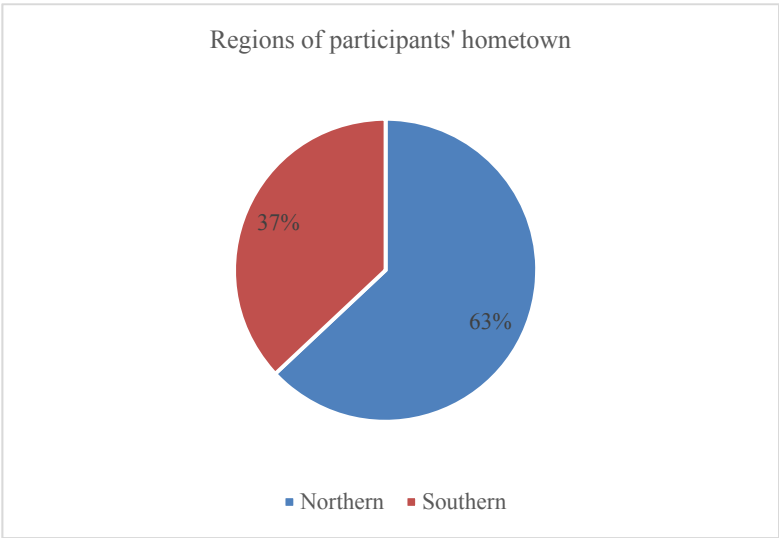
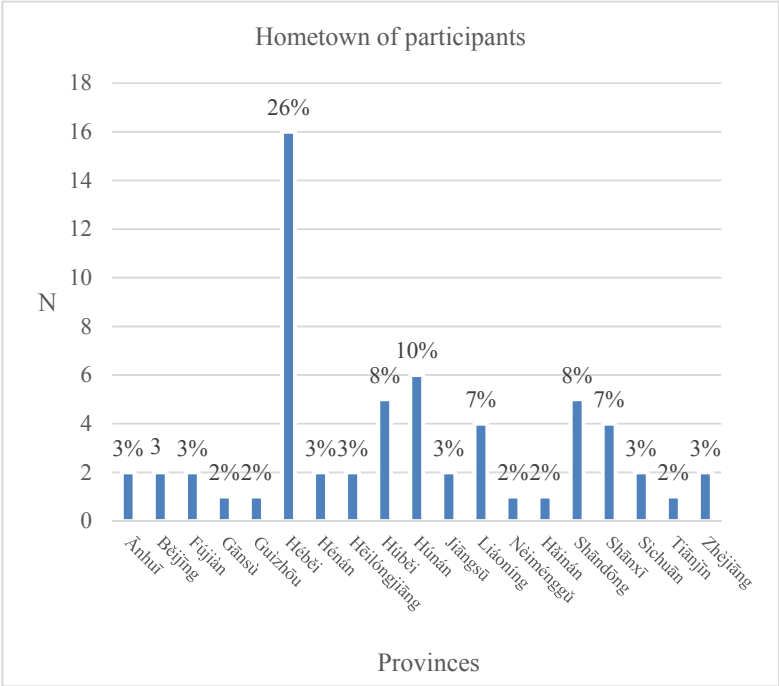
(1) Gender:

Gender	N	Percentage
Male	16	26.23%
Female	45	73.77%
Total N	61	

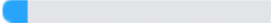
(2) Age ranges:

Age ranges	N	Percentage
Under 18	0	0%
18~25	26	42.62%
26~30	18	29.51%
31~40	16	26.23%
41~50	1	1.64%
51~60	0	0%
60 and above	0	0%
Total N	61	

(3) Hometown:



(4) Education background:

Degrees	N	Percentage
B.A.	6	 9.84%
Postgraduates	36	 59.02%
Ph.D	19	 31.15%
Total N	61	

B. Participants' judgments on the test sentences (re-categorized into three groups)

(1) **Group 1:** [VP [adv *zhèngzài*] [VP [PP *zài* NP] VP]]

a. 这群人正在在我家打边炉。

Zhè qún rén zhèngzài zài wǒ jiā dǎ biānlú.

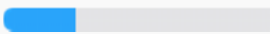
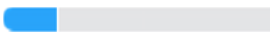
this CL people just at+PROGR 1SG home strike hot-pot

‘This group of people is just enjoying a Cantonese hotpot at my place.’

Acceptability scale	N	Percentage
Completely unacceptable	32	 52.46%
Probably unacceptable	19	 31.15%
Probably acceptable	6	 9.84%
Completely acceptable	3	 4.92%
I don't know	1	 1.64%
Total N	61	

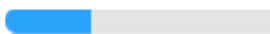
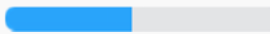
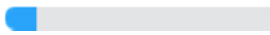
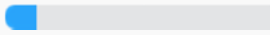
b. 比赛正在在兰州举行。

Bìsài zhèngzài zài Lánzhōu jǔxíng.
 competition just PROGR+at Lanzhou hold
 ‘The match is just going on in Lanzhou.’

Acceptability scale	N	Percentage
Completely unacceptable	31	 50.82%
Probably unacceptable	16	 26.23%
Probably acceptable	12	 19.67%
Completely acceptable	1	 1.64%
I don't know	1	 1.64%
Total N	61	

c. 这部电视剧正在在北京拍摄。

Zhè bù diànshìjù zhèngzài zài Běijīng pāishè.
 this cl TV.drama just PROGR+at Beijing shoot
 ‘This TV drama is just being filmed in Beijing.’

Acceptability scale	N	Percentage
Completely unacceptable	19	 31.15%
Probably unacceptable	28	 45.9%
Probably acceptable	7	 11.48%
Completely acceptable	7	 11.48%
I don't know	0	 0%
Total N	61	

d. 他正在在和朋友交谈。

Tā [AspP zhèngzài [AspP zài [PP hé péngyǒu] jiāotán]]].

3sg just PROGR with friend chat

‘He is just chatting with friends.’

Acceptability scale	N	Percentage
Completely unacceptable	40	65.57%
Probably unacceptable	19	31.15%
Probably acceptable	2	3.28%
Completely acceptable	0	0%
I don’t know	0	0%
Total N	61	

(2) Group 2: [adv zhèngzài] [asp zài] VP

a. 她正在在学习中文。

Tā zhèngzài zài xuéxí zhōngwén.

3sg just PROGR learn Chinese

‘She is just learning Chinese.’

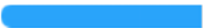
Acceptability scale	N	Percentage
Completely unacceptable	46	75.41%
Probably unacceptable	14	22.95%
Probably acceptable	1	1.64%
Completely acceptable	0	0%
I don’t know	0	0%
Total N	61	

b. 同事们正在在买礼品。

Tóngshìmen zhèngzài zài gòumǎi lǐpǐn.

colleagues just PROGR buy gift

‘The colleagues are just buying gifts.’

Acceptability scale	N	Percentage
Completely unacceptable	44	 72.13%
Probably unacceptable	13	 21.31%
Probably acceptable	0	 0%
Completely acceptable	4	 6.56%
I don't know	0	 0%
Total N	61	

c. 积雪正在在迅速地旋转着。

Jīxuě zhèngzài zài xùnsùde xuánzhuǎn-zhe.

snow just PROGR rapidly spin -dur

‘The snow is just spinning rapidly.’

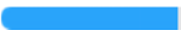
Acceptability scale	N	Percentage
Completely unacceptable	42	 68.85%
Probably unacceptable	17	 27.87%
Probably acceptable	2	 3.28%
Completely acceptable	0	 0%
I don't know	0	 0%
Total N	61	

d. 渔业资源正在在急剧减少。

Yúyè zīyuán zhèngzài zài jíjù jiǎnshǎo.

fishery resource just PROGR sharp decrease

‘Fishery resources are just declining sharply.’

Acceptability scale	N	Percentage
Completely unacceptable	39	 63.93%
Probably unacceptable	20	 32.79%
Probably acceptable	2	 3.28%
Completely acceptable	0	 0%
I don't know	0	 0%
Total N	61	

(3) Group 3: [_{adv} zhèngzài] [_{VP} [_{Verb} zài] [_{PostP} NP postposition]]

a. 各方保证正在在会谈中。

Gè fāng bǎozhèng zhèngzài zài huìtán zhōng

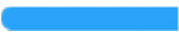
each side assure just be.at talks in

‘All parties assure to be presently (engaged) in talks.’

Acceptability scale	N	Percentage
Completely unacceptable	40	 65.57%
Probably unacceptable	18	 29.51%
Probably acceptable	3	 4.92%
Completely acceptable	0	 0%
I don't know	0	 0%
Total N	61	

b. 他似乎此刻正在在冷风里。

Tā sìhū cǐkè zhèngzài zài lěng fēng lǐ.
 3sg seemingly at.present just be.at cold wind in
 'It seems as though he is just in a cold wind right now.'

Acceptability scale	N	Percentage
Completely unacceptable	39	 63.93%
Probably unacceptable	15	 24.59%
Probably acceptable	5	 8.2%
Completely acceptable	2	 3.28%
I don't know	0	 0%
Total N	61	

c. 本公司的‘蓝星大厦’正在在建设中。

Běn gōngsī de 'lán-xīng-dàshà' zhèngzài zài jiànshè zhōng.
 1PL company SUB blue-star-building just be.at build in
 'The *blue star* building of our company is just under construction.'

Acceptability scale	N	Percentage
Completely unacceptable	32	 52.46%
Probably unacceptable	16	 26.23%
Probably acceptable	4	 6.56%
Completely acceptable	9	 14.75%
I don't know	0	 0%
Total N	61	

d. 电话正在在接通中。

Diànhuà zhèngzài zài jiētōng zhōng.
 phone just be.at put.through in
 ‘The phone is just connecting.’

Acceptability scale	N	Percentage
Completely unacceptable	39	63.93%
Probably unacceptable	18	29.51%
Probably acceptable	2	3.28%
Completely acceptable	2	3.28%
I don't know	0	0%
Total N	61	

d. 电话正在在接通中。

Diànhuà zhèngzài zài jiētōng zhōng.
 phone just be.at put.through in
 ‘The phone is just connecting.’

Acceptability scale	N	Percentage
Completely unacceptable	39	63.93%
Probably unacceptable	18	29.51%
Probably acceptable	2	3.28%
Completely acceptable	2	3.28%
I don't know	0	0%
Total N	61	

